

Contingent Molecular Coding in a Specified Reaction Architecture: Formation, Causal Heredity, and Output-Mediated Maintenance

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Abstract

The formation of a molecular assignment and the origin of the capabilities that implement it are distinct explanatory problems. This paper studies assignment formation within a specified recognition, assembly, templating, and readout architecture whose initial state contains no cross-domain carrier or privileged mapping. The principal result is a joint existence theorem: one fixed, independently assignment-symmetric reaction-compartment grammar supports contingent readout, topology-dependent causal heredity, and output-mediated maintenance on a common admissible parameter regime, without a mapping-quality reward or an externally activated reaction stage. Local assembly and nonlinear reinforcement generate two symmetry-related organizations that mediate opposite symbol-conditioned channels. Matched interventions distinguish the channel’s dependence on its carrier, the effect of inherited pairing topology on descendant channels, and the contribution of decoded outputs to future interpreter capacity. An elementary fuel-driven mass-action realization retains these properties under the stated scale separation. Genesis has strictly positive probability; high-probability lineage retention is established separately from constructed exact-balanced mature parents, over horizons growing exponentially with compartment scale. Finite-system reset and local free-rider bounds delimit that retention. The contribution is a compatible construction and a set of quantitative causal discriminators for a restricted operational molecular code. It does not derive the molecular origin of the supplied capabilities, mapping-sensitive interpreter self-construction, a high-probability complete origin history, or a historical pathway to the Standard Genetic Code. The construction makes precise what remains to be explained when a model generates a coding relation within an already specified architecture.

Keywords: molecular coding; chemical reaction networks; causal heredity; matched interventions; output-mediated maintenance; stochastic compartments; symmetry breaking; origin of interpretation

1. Introduction: assignment formation and the origin of interpretation

The originating question is how a physical system first acquires the capacity to interpret molecular sequences: where does a decoder come from? Producing a sequence or selecting a correspondence does not, by itself, explain the capacities that make the correspondence functional. Molecular-code formalisms accordingly distinguish a set of potential mappings from the material context that implements a particular mapping [Görlisch and Dittrich 2013], while interpreter-origin research asks how such functional organization can itself arise [Wills 2009].

Two explanatory tasks must therefore be separated. One is the formation and maintenance of a particular assignment within a specified reaction architecture. The other is the formation of the recognition, assembly, copying, and readout capabilities constituting that architecture. This paper addresses the first task. Its model supplies domain-local recognition and actuation, composable receptor-effector readout, and homotypic relational templating. These are substantive assumptions whose molecular origin is not derived. The initial state lacks an assembled cross-domain carrier, and no particular assignment or mapping-quality reward is supplied.

Within those assumptions, a nontrivial question remains: can fixed local dynamics generate alternative assignments, transmit a realized assignment through physical lineage organization, and make the channel’s outputs contribute to its future operation? Separate examples of switching, self-support, or persistence do not establish their compatibility in a shared model and parameter regime. Nor does passive correlation identify the physical relation responsible for a descendant channel. The present construction addresses these joint dynamical and causal requirements.

Autocatalytic organization, hypercycles, RAF sets, and composition-based inheritance have a substantial body of prior work [Eigen and Schuster 1979; Segré, Ben-Eli and Lancet 2000; Vasas, Szathmáry and Santos 2010; Hordijk and Steel 2015]. Their existence motivates a more specific target here: an intervention-defined, remappable channel whose realized pairing topology contributes causally to heredity and whose

outputs maintain subsequent readout capacity. Establishing this target is a conditional contribution to the broader origin problem, not a solution to the origin of every required molecular capability.

This question sits within a substantial prior literature. Assignment-catalyst and self-description models already addressed how coding relations and their catalysts might co-develop within template-product systems [Bedian 1982, 2001]. Nieselt-Struwe and Wills gave an early dynamical account of the emergence of genetic coding in a physical system [Nieselt-Struwe and Wills 1997]. Wills subsequently developed models and conceptual analyses of self-organizing, physical, and reflexive genetic-code interpreters, including spontaneous mutual ordering of nucleic acids and proteins [Wills 1993, 2009, 2014a, 2014b, 2016, 2023], while Füchslin and McCaskill studied evolutionary self-organization and stabilization of cell-free genetic coding [Füchslin and McCaskill 2001]. Wills, Nieselt and McCaskill formulated coding specificity as a physico-informatic problem [Wills, Nieselt and McCaskill 2015], and Carter and Wills emphasized the interdependence and reflexivity of assignment catalysts and translation, including the bootstrap difficulty for an RNA-only origin [Carter and Wills 2018; Wills and Carter 2018; Carter 2024]. These works preclude any broad claim that coding emergence, assignment catalysis, physical interpretation, or reflexivity is introduced here.

Other frameworks begin with a coding channel or a set of assignments and study its organization under an information-theoretic or evolutionary objective [Tlusty 2007, 2008; Piraveenan, Polani and Prokopenko 2007]. Molecular-code capacity and contingent mappings have been formalized directly in chemical reaction networks and other physical media [Görlich and Dittrich 2013; Dittrich 2018]. Symbol-matter and semantic-closure traditions analyze how materially embodied relations acquire or sustain interpretive function [Rocha 1998; Pattee 2007; Sharov 2009b], and semantic closure has been demonstrated in an artificial chemistry containing an executable constructor architecture [Clark, Hickinbotham and Stepney 2017]. Sharov proposed a coenzyme autocatalytic model linking primitive coding elements, encoded functions, compartment propagation, and heredity [Sharov 2009a]. Bernhardt and Tate proposed a transition from noncoded to coded protein synthesis in an RNA-world setting [Bernhardt and Tate 2010]. Chemical-historical and simulation studies have also proposed stepwise routes from RNA-peptide mutualism or signaling to translation and coding [Vitas and Dobovišek 2018; Chatterjee and Yadav 2019; Spirov 2023]. Sequence-based reconstruction has separately proposed an ordered history of tRNA_{ome}, aminoacyl-tRNA-synthetase, and Standard Genetic Code development [Lei, de Farias and Burton 2026]. These studies address different starting architectures and explanatory targets; comparison with the present construction must distinguish those assumptions from the outcomes demonstrated. More recently, Nunes Palmeira et al. modeled the emergence of genetic heredity in dividing protocells under growth selection, with copying and rudimentary translation included as model processes [Nunes Palmeira et al. 2026]. Deacon, Garcia-Valdecasas and Monzón-Marqués proposed an autogenic route to molecular information but expressly did not attempt to explain the genetic code [Deacon et al. 2026], while Segura formulated a categorical account of code-bearing autopoietic and reflexive closure [Segura 2026, preprint].

The principal contribution is a **joint existence theorem with matched causal identification**. One fixed local reaction grammar combines independently relabelable molecular domains, locally assembled relation topology, opposite remappable channels, parent-relative causal heredity, and output-mediated support of future interpreter capacity. A low-molecularity mass-action realization and a common relatively open parameter neighborhood make this a compatibility result for the coupled system. The result does not follow merely from observing the ingredients separately: coupling changes the resource balances and stability conditions, and the heredity comparison requires an explicit matched erasure protocol.

Its distinctive value lies in this explicit conjunction, the causal tests attached to it, and the quantitative boundaries that prevent stronger interpretations. Coding self-organization, assignment catalysts, reflexivity, and interpreter-origin questions are established topics, so none is claimed as a new general idea here. The literature comparison in Section 10.6 and Appendix J identifies the particular objects being compared; it does not establish an exhaustive priority claim.

The word *code* itself is not used uniformly across this literature. Noisy-channel models commonly begin with a coding map and optimize its accuracy, diversity, or cost [Tlusty 2007, 2008]; molecular-code and semantic-capacity formalisms emphasize contingent mappings that a physical medium can implement [Görlich and Dittrich 2013; Dittrich 2018]; philosophical analyses distinguish several senses of the genetic code's arbitrariness [Stegmann 2004]; Code Biology emphasizes adaptor-mediated relations and a lower coding threshold [Lacková and Faltýnek 2021; Barbieri 2025]; and semantic-closure accounts place additional weight on an interpretation system that sustains the relation between code and function [Rocha 1998; Pattee 2007; Sharov 2009b; Wills 2009]. These are not equivalent criteria. To prevent terminological

disagreement from carrying the mathematical burden, the neutral theorem object in this paper is a **contingent, remappable, causally mediated cross-domain channel**. The term *operational molecular code* refers to that object under the intervention and specificity criteria in Section 2.3; causal heredity and output-mediated maintenance are assessed separately in Section 2.4. Stronger claims about full biological interpretation, mapping-sensitive self-construction, or semiotic meaning are not inferred from the theorem.

The construction is designed to meet the logical requirements of the existence theorem; it is not claimed to minimize reaction count, molecular engineering complexity, or prebiotic difficulty. Candidate symbols and outputs are binary. Domain-local receptor and effector capabilities are primitive physical properties; what is absent initially is a cross-domain relation equivalent to a decoder or assignment table. Generic receptor–effector relations nucleate without a preferred pairing. A local cooperative rule reinforces already formed relation topology. In the large-system limit this produces two symmetry-related broken states. The realized relation then mediates a symbol-conditioned output channel. Outputs are transduced to a common maintenance currency that repairs damaged relation complexes, closing a causal loop without introducing a post hoc stage switch. The effective nonlinear reactions are then embedded into a low-molecularity, explicitly fuel-driven mass-action network. Finally, compartments grow neutrally and divide by unbiased molecular partition, allowing the causal inheritance of the physical relation topology to be tested against a matched topology-erasure control.

Five aspects of the analysis are essential. First, coding is defined by intervention and mediation, not by passive correlation. Second, the reaction dynamics are specified independently of the code assay; mapping statistics never enter reaction rates. Third, negative results delimit the construction: identity-erasing dynamics, linear or concave reinforcement, closed finite fuel, passive finite ancestral memory, permanent finite-system fidelity under recurrent reset, and an ungated causal fingerprint are all insufficient. Fourth, stochastic symmetry breaking is treated with the correct order of limits: finite systems choose one of two equivalent coding basins, while an exactly symmetric deterministic fluid trajectory does not choose a branch at fixed time. Fifth, the final theorem is explicitly a model-existence result. It does not identify a historical pathway on early Earth, demonstrate prebiotic availability of the abstract modules, or explain the origin of the Standard Genetic Code.

Four contributions organize the analysis:

1. **Compatible dynamics in one specified architecture.** Sections 4–8 construct assignment formation, readout, output-mediated maintenance, and conditional lineage retention in the same elementary reaction family. The Main Theorem supplies a common admissible parameter regime with the stated order of choices.
2. **A causal test of inherited assignment.** Theorems 11, 15, and 18 compare intact and composition-matched, topology-erased parents. This distinguishes a descendant channel’s dependence on the inherited pairing relation from persistence of molecular abundance alone; the exact comparison is restricted to the declared balanced-parent and assay conditions.
3. **Separate identification of three functions.** Theorem 8 and Section 9 distinguish carrier-mediated readout, topology-dependent heredity, and output-mediated maintenance. Theorem 19 states calibrated mismatch bounds under which the latter two interventions separate the specified comparator classes after the operational channel test passes.
4. **Quantitative limits and tradeoffs.** Theorems 3, 5–6, 9, 13, 16, and 17 delimit reinforcement, stochastic selection, energy supply, ancestral memory, finite-system fidelity, and a declared local free-rider family. Section 9.2 links coding specificity to the reduced family’s invasion boundary.

These are claims about the defined model and comparators. Their relevance to the originating question is to isolate which coding functions can arise once specified capabilities are available, and to expose the further mechanistic explanation that the capabilities themselves require. Figure 1 locates both the constructed sequence and its starting assumptions.

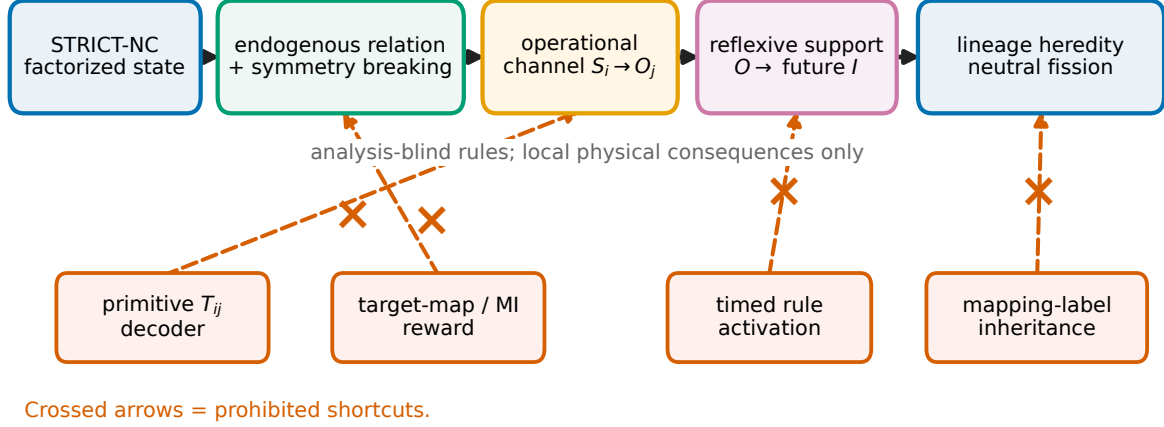


Figure 1: **Research object and forbidden shortcuts.** The admitted construction proceeds from a STRICT-NC factorized state to endogenous cross-domain relation formation, symmetry-broken relation topology, an operational symbol-conditioned channel, reflexive maintenance, and lineage heredity. Recognition, actuation, composable readout, and homotypic templating are supplied capabilities throughout; the diagram does not depict their origin. Crossed side arrows mark prohibited shortcuts within this architecture: a primitive decoder-equivalent T_{ij} , target-map/MI reward, externally timed post-code rule activation, and inheritance of an analysis-side mapping label.

2. Operational definitions and no-hidden-decoder discipline

2.1 Physical model and analysis layer

A candidate physical model is

$$\mathcal{M} = (\mathcal{S}, \mathcal{R}, N, a, \mu_0, \mathcal{B}, \mathcal{E}, \mathcal{L}),$$

where $\mathcal{S}$ is a set of physical species or material states, $\mathcal{R}$ is a set of reactions or events, N is the stoichiometric update matrix, $a_r(x)$ is the local propensity of event r , μ_0 is the initial state or law, $\mathcal{B}$ specifies boundary resources and chemostats, $\mathcal{E}$ records the declared energetic abstraction, and $\mathcal{L}$ is an optional compartment/lineage process. We use the standard well-mixed stochastic chemical-kinetics convention and its volume-scaled deterministic relation [Gillespie 1977; Kurtz 1972]. A stochastic CRN has generator

$$(\mathcal{G}f)(x) = \sum_{r \in \mathcal{R}} a_r(x) [f(x + N_r) - f(x)].$$

The observer separately defines an assay interface

$$\mathcal{A} = (\Sigma, \Omega, \phi_\Sigma, \phi_\Omega, \tau),$$

where Σ and Ω are candidate symbol and output classes. The physical generator is specified before $\mathcal{A}$ is applied. Removing the assay, fitted mapping, mutual-information calculation, or any code-quality statistic leaves the physical path law unchanged.

This separation is the first anti-circularity requirement: the chemistry cannot query the analyst’s interpretation of it.

2.2 STRICT-NC baseline

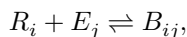
Let G_Σ and G_Ω be descriptor-preserving permutations of the candidate symbol and output classes. The baseline is **strictly assignment-neutral** when the dynamics and initial law are equivariant/invariant under the independent action

$$G_{\Sigma} \times G_{\Omega},$$

rather than only under a diagonal relabeling that preserves a preassigned pair.

A primitive type or rule is **decoder-equivalent** if its fixed presence is sufficient, without de novo organization, to mediate a reproducible nontrivial symbol-dependent output choice. Supplying all possible pair-specific decoders symmetrically does not evade this condition: a primitive family T_{ij} whose rule semantics are “read S_i and produce O_j ” already contains the cross-domain assignment architecture.

The reaction grammar is typed. A symbol-side primitive may expose or test only a symbol/receptor port i , and an output-side primitive may expose or actuate only an effector/output port j . The sole operation allowed to create a state carrying both ports from separate primitive objects is the uniform Cartesian-product assembly schema



whose kinetic form is identical for every (i, j) . The subscript on B_{ij} records the physical provenance of the two components; it is not a primitive lookup entry. After this assembly, a rule may retain both indices only because one of its reactants is already a constructed carrier bearing both ports. A challenge rule may inspect only that carrier’s receptor port, and an actuation rule may expose only its effector port. A direct rule $S_i \rightarrow O_j$, a primitive T_{ij} , or any rate or branch selected by the ordered pair (i, j) is prohibited.

We call $(\mathcal{M}, \mu_0)$ a **STRICT-NC** baseline for an assay interface if: (i) the dynamics are analysis-blind; (ii) independent assignment neutrality holds; (iii) the preceding typed-grammar condition holds and no decoder-equivalent primitive is initially present; (iv) no active cross-domain interpreter is present at the origin; (v) no mapping-level fitness or target score enters any rule; and (vi) material/resource/energy boundaries are declared.

Scope of “decoder-free.” STRICT-NC does not mean that the model contains no domain-local recognition or actuation capabilities. The construction takes symbol/receptor recognition within the symbol-side domain and effector/output actuation within the output-side domain as primitive physical capabilities. What is absent initially is a **pre-existing cross-domain decoder-equivalent relation** joining those domains. Formally, independent permutations of the symbol/receptor and effector/output domains act equivariantly on every reaction schema and leave the STRICT-NC initial state invariant; therefore the grammar contains no privileged cross-domain bijection. The theorem concerns genesis of a contingent cross-domain carrier topology and the channel it subsequently mediates, not genesis of all molecular recognition from featureless chemistry.

The construction below assumes a composable receptor–effector readout architecture and homotypic relational templating. These general capabilities are specified by the reaction grammar; their molecular origin is not derived. What is generated endogenously is an instantiated cross-domain relation and its realized orientation. Thus STRICT-NC is a condition on the declared model and starting state, not a certificate that the architecture contains no functionally specialized assumptions. Independent relabeling excludes a privileged assignment; it does not derive the recognition, templating, or actuation mechanisms.

2.3 Operational channel and coding specificity

Under a matched intervention $do(\sigma)$, let $J_t^r(\omega \mid \sigma)$ be expected output-event flux over the assay window. The $do(\cdot)$ notation follows the intervention–observation distinction in causal modeling [Pearl 2009]; here every intervention is defined directly as a physical modification of the model or assay protocol, so no causal effect is inferred from observational conditioning alone. When total output flux is positive, define

$$K_t^r(\omega \mid \sigma) = \frac{J_t^r(\omega \mid \sigma)}{\sum_{\omega'} J_t^r(\omega' \mid \sigma)}.$$

The assignment-free causal specificity is

$$C(K) = \max_{\sigma \neq \sigma'} \text{TV}(K(\cdot \mid \sigma), K(\cdot \mid \sigma')).$$

For a binary row-stochastic channel,

$$C(K) = |\det K|.$$

The determinant sign is an orientation coordinate; its magnitude is invariant under independent relabeling of the two domains. Mutual information may be reported post hoc but is never used as a physical objective.

An endogenous subsystem I counts as an interpreter only if the channel is reproducible with I intact and an ablation matched in the declared formal moiety totals, resource boundary activities, and sample load demonstrates mediation. Equal free energy or intervention work is not established by this matching. The normalized channel is not assigned an artificial value when total output flux vanishes. Instead, mediation is established if either (i) the matched ablation destroys the required channel by making the total output flux zero for at least one required challenge, or (ii) the ablated channel remains defined for all required challenges and strictly lowers $C(K)$ or the declared assignment-fidelity observable. A direct stoichiometric transformation $S_i \rightarrow O_i$ whose output identity is fixed by the consumed symbol is not, by itself, an operational code. Counterfactual remappability—two reachable interpreter states realizing opposite mappings on the same symbol/output classes—is a strong sufficient discriminator.

Controlled readout protocol. All state-to-channel identifications for the constructed model use the following finite-window assay on a separate sample at its current volume. The lineage itself continues with the full autonomous chemistry. In the sample, suspend growth, imports, and all reactions except the symbol-binding, symbol-release, and output-producing reactions (32). Before applying either challenge, release every bound symbol by the same preparation operation $U_{ij} \mapsto A_{ij} + S_i$, and set the external symbol activities to $S_i = S_* > 0$, $S_{1-i} = 0$ for challenge i . Hold $Q = Q_* > 0$. Count only new firings of the output-producing reaction during a fixed $\tau > 0$; inherited O_j is held inactive and is not counted. The preparation preserves each existing receptor-effector pairing and all formal moieties, with released symbols accounted for at the boundary. It creates no new pairing, selects no target assignment, and acts identically on all symmetry-related carrier classes. Only the challenged receptor port differs between the two arms.

Let $L_{ij} = A_{ij} + U_{ij}$ denote the accessible carrier concentration **before** preparation. The same quantity is the free carrier concentration just after preparation. Template-bound carriers in H, X, Y , damaged carriers, and parasite classes remain inactive during this readout measurement. In the elementary scaling of Appendix E set

$$a_S = \frac{k_S^+ S_*}{\eta}, \quad b_S = \frac{k_S^-}{\eta^2}, \quad c_Q = \frac{k_Q Q_*}{\eta}, \quad \lambda_S = a_S + b_S + c_Q.$$

For a sample of volume V , the exact expected new output count is

$$J_i^\tau(j) = V L_{ij} \Gamma_{\eta, \tau},$$

$$\Gamma_{\eta, \tau} = \frac{c_Q a_S}{\lambda_S} \left[\tau - \frac{1 - e^{-\lambda_S \tau}}{\lambda_S} \right] > 0.$$

This identity holds both for deterministic integrated flux and for the finite-molecule stochastic expectation, conditional on the sampled state (Appendix A). Thus, whenever each row total $L_i = \sum_j L_{ij}$ is positive,

$$\mathcal{K}(x)(j | i) = \frac{L_{ij}}{L_i}.$$

The same map applies exactly to samples of volumes $N, 2N, 4N$; there is no interchange of a short-time, fast-reaction, or large-volume limit. In the reduced relation model the analogous controlled readout keeps x_{ij} fixed and has a common per-carrier output intensity, yielding $K(j | i) = x_{ij} / \sum_j x_{ij}$. The channel measures available carrier-mediated readout under the declared controls. It is not asserted to equal the transient channel obtained by changing one symbol while assembly, maintenance, and growth continue. The matched support interventions of Section 5 use that full dynamical system and are separate experiments.

For interpreter ablation, apply the same preparation and convert every accessible A_{ij} in the ablated sample to a receptor-port-inactive isomer A_{ij}^0 , with identical formal composition and load. The intact

sample retains functional A_{ij} . The inactive isomer cannot bind a challenge; all other sample controls and boundary activities are identical. With no inherited bound symbol and all assembly routes suspended, every required new output flux in the ablated sample is zero. No normalized ablated channel is assigned. This is an explicit idealized intervention on the sample, not an additional autonomous reaction or a claim that a particular laboratory reagent realizes it.

2.4 Heredity, persistence and reflexive closure

For two channels on the same finite challenge and output alphabets, define the rowwise total-variation metric

$$d_K(K, L) = \max_{\sigma \in \Sigma} \text{TV}(K(\cdot \mid \sigma), L(\cdot \mid \sigma)).$$

It is used only when every required row in both channels has positive total output flux. A channel is heritable only if a parent–descendant d_K criterion is satisfied **and** a matched parental intervention changes the descendant channel distribution. Common-environment re-formation of a similar state is therefore not sufficient.

An interpreter is reflexively maintained when the following two causal directions are independently established:

$$I \longrightarrow \text{code-mediated output}, \quad \text{code-mediated output} \longrightarrow \text{future } I \text{ capacity}.$$

The second edge requires a matched intervention that blocks the supporting output while preserving the instantaneous physical state and matched burden as far as the model permits. Here **matched burden** has the narrow physical meaning used by the stated intervention: equality of the declared receptor, effector, symbol/readout, and fuel moiety totals; equality of structural-class totals and non-pair bulk coordinates when the topology-erasure kernel is used; and the same challenge schedule, volume protocol, and observation time. It does not mean equality of every microscopic chemical identity, because the intended intervention must alter the targeted causal route.

2.5 Joint theorem conditions

The main theorem will establish the following eight clauses on a nonempty open symmetry-respecting parameter region and a compact interior core (Table 1).

Table 1 — Joint conditions C0–C7.

Clause	Requirement
C0	STRICT-NC origin with explicit material/energy boundary
C1	positive-probability endogenous readout/interpreter genesis
C2	operational, remappable code rather than direct chemistry
C3	explicit local causal consequence with no mapping-level fitness
C4	persistence through nonzero stochastic turnover
C5	parent-relative causal heredity across at least two lineage events
C6	reflexive interpreter maintenance by code-mediated output
C7	coexistence on a nonempty open parameter region, with quantitative robustness

The target is intentionally not a claim about triplet coding, aminoacyl-tRNA synthetases, ribosomes, or the historical evolution of the Standard Genetic Code.

2.6 Terminological scope: initial wiring, code, and reflexivity

The mathematical claims are easiest to assess if three disputed terms are separated from one another.

Box 1 — Established properties and interpretive scope.

Statement	Conclusion in this analysis
No pre-existing cross-domain assignment relation or decoder-equivalent lookup table	Established model property: no pair-bearing carrier is initially present; the typed grammar requires de novo assembly before cross-domain readout, and independent relabeling excludes a privileged assignment
No domain-local recognition or actuation primitives of any kind	Not claimed: symbol docking and output-side actuation are primitive physical capabilities
Endogenous contingent and remappable symbol-conditioned channel	Proved under the operational intervention criteria
The object satisfies every proposed biological or semiotic definition of “code”	Not claimed: the mathematical result does not depend on adopting a particular biological terminology; “operational code” refers to the stated criteria
Code-mediated output causally supports future interpreter capacity	Proved as output-mediated reflexive maintenance
Mapping-specific outputs construct the interpreter required by that mapping	Not proved; this is a stronger reflexive-closure target
A high-probability historical path from arbitrary non-coded chemistry to a modern translation apparatus	Not claimed

The expression *decoder-free* is used only in the restricted initial-state sense specified in Section 2.2: pair-bearing carriers are absent initially, their construction follows the typed grammar, and no privileged cross-domain relation $i \leftrightarrow j$ is supplied. The architecture still provides discriminatory binding, templating, and actuation. Independent relabeling and initial carrier absence are distinct checks, and neither establishes the molecular origin of these capabilities.

Second, the proved object is stronger than an ordinary fixed molecular switch but weaker than a complete modern genetic decoder. A switch can map a control variable to one of two states through a fixed direct mechanism. Here the same domain-local primitives support two physically equivalent, opposite cross-domain assignments; the assignment is carried by a removable relation topology; relation ablation destroys or weakens the channel; and that topology has a parent-relative causal effect on descendant channels. These properties motivate the operational-code terminology. Readers who prefer a vocabulary-free formulation may substitute **contingent remappable molecular transducer** everywhere without changing any theorem.

The distinction also locates the result relative to semantic-capacity and lower-threshold proposals [Dittrich 2018; Lacková and Faltýnek 2021; Barbieri 2025]. The present theorem establishes contingency, remappability, mediation, repetition across lineages, and a restricted form of reflexive support. It does not claim to settle representation, historicity, interpretation, or meaning in every broader philosophical sense.

3. Boundary theorems: what cannot count as coding

The construction is easiest to understand by first isolating mechanisms that fail.

Theorem 1 — identity-erasure null

Let $\mathcal{G}_0$ and $\mathcal{G}_1$ be the generators under two matched symbol challenges. Suppose there is an output-sufficient projection π , a **common** quotient generator $\bar{\mathcal{G}}$, and quotient output-mark intensities $\bar{\lambda}_\omega$ such that, for every bounded quotient test function f [Buchholz 1994],

$$\mathcal{G}_i(f \circ \pi) = (\bar{\mathcal{G}}f) \circ \pi, \quad \lambda_{i,\omega}(x) = \bar{\lambda}_\omega(\pi x), \quad i \in \{0, 1\}.$$

Assume also that the common quotient martingale problem is well posed and nonexplosive, that the two challenges induce the same projected initial law, and that every expected output count in the finite assay window is finite, with a positive total in every required channel row. Then the expected marked-output counts agree under the two interventions. Consequently the normalized intervention-channel rows are identical and

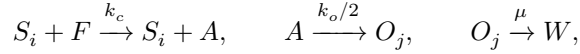
$$\boxed{C(K) = 0.}$$

Proof. Well posedness and equality of the projected initial laws give the same quotient process law. For every output label, the counting-process compensator identity gives

$$E_i N_\omega(t, t + \tau) = E_i \int_t^{t+\tau} \bar{\lambda}_\omega(\pi X_s) ds.$$

The right-hand side is the same in both challenges. Finite positive row totals therefore give equal normalized rows. Equality of the joint marked path laws is not needed and does not follow merely from equality of the quotient generator and marginal mark intensities. $\square$

A concrete null CRN illustrates the point:



with identical rates for $i, j \in \{0, 1\}$. Symbols genuinely alter the total activation flux, yet they do not carry causal information about relative output identity. Local causal activity is therefore insufficient for coding.

Corollary 1 — output-relevant identity memory is necessary under the null hypotheses

If $C(K) > 0$, no output-sufficient strongly lumpable quotient that erases the active symbol distinction can remain valid over the assay window. Some physical state or history relevant to output selection must carry the challenged identity across the causal gap.

This is conditional on the quotient hypotheses; it is not a universal theorem that all possible codes require a literal receptor–effector bond.

Lemma 1 — single-domain memory is insufficient under unbroken output symmetry

Fix a matched symbol challenge. Suppose a mediator preserves symbol identity, the output-side transition law is invariant under a group acting transitively on output labels, the conditional initial law after that challenge is invariant under the same output action, and the matched intervention/boundary conditions are output-equivariant. Then the output law is invariant under the transitive output action and therefore cannot produce a nonuniform relative output response. A successful minimal construction must therefore create an output-relevant **cross-domain relational state**, not merely a one-domain memory variable.

Theorem 2 — minimal symmetric relation reachability

Take exactly one molecular object of each receptor type R_0, R_1 and each effector type E_0, E_1 , initially unpaired. At this scaffold stage, allow only irreversible generic monovalent relation assembly: any free R_i may bind any free E_j at the same rate $k_a > 0$. There are exactly two perfect matchings,

$$M_+ = \{(0, 0), (1, 1)\}, \quad M_- = \{(0, 1), (1, 0)\}.$$

From the empty state, all four first edges have equal propensity. Each first edge uniquely determines the remaining compatible edge, hence

$$\boxed{\Pr(M_+) = \Pr(M_-) = \frac{1}{2}.}$$

If a challenged S_i activates whichever effector is physically linked to R_i , then

$$K_+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus each reached organization has $C(K) = 1$, the same symbol can produce the opposite output in the opposite relation topology, and ablation of the cross-domain bonds eliminates the gated output route. The relation topology therefore carries the assignment without an initial pair-indexed decoder.

This scaffold is a reachability result, not yet a persistent chemical origin mechanism.

4. Endogenous relation formation and operational coding

4.1 Reversible turnover and why stationary signed mapping is the wrong order parameter

Adding unbiased dissociation to the single-copy 2×2 matching scaffold of Theorem 2 produces a finite irreducible CTMC. Its stationary law is exactly symmetric under output relabeling. Therefore the infinitely long stationary intervention channel has identical rows and $C(K_\infty) = 0$, even though the chain spends a positive fraction of time in perfectly matched configurations.

For association rate k_a , dissociation rate k_d , and $z = k_a/k_d$, the exact seven-state partition function is

$$Z = 1 + 4z + 2z^2,$$

with total perfect-matching occupancy

$$Q_{\text{struct}} = \frac{2z^2}{1 + 4z + 2z^2} > 0.$$

Finite structural matching occupancy is not itself a coding phase, and $C(EK)$ is not an admissible order parameter for a symmetric mixture of opposite codes. Specificity must be measured within a realized organization or lineage before averaging over orientations.

4.2 Local reinforcement model

Let x_{ij} be relation-complex densities and

$$r_i = 1 - \sum_j x_{ij}, \quad e_j = 1 - \sum_i x_{ij}.$$

Let $g \in C^1([0, 1])$ satisfy $a + g(x) \geq 0$ for every $x \in [0, 1]$, with $a + g(0) > 0$. Consider the assignment-neutral shared-resource family

$$\dot{x}_{ij} = [a + g(x_{ij})]r_i e_j - dx_{ij}, \tag{1}$$

where $a, d > 0$ and the same local reinforcement law g applies to every formed topology.

At a fully symmetric equilibrium $x_{ij} = x$, with $s = 1 - 2x$,

$$[a + g(x)]s^2 = dx. \tag{2}$$

A checkerboard perturbation

$$\delta X = \epsilon \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

has zero row and column sums, so to first order it does not alter the free resources. Its eigenvalue is

$$\lambda_A = g'(x)s^2 - d = \frac{s^2}{x} [xg'(x) - g(x) - a]. \tag{3}$$

Theorem 3 — reinforcement curvature criterion

The checkerboard eigenvalue of the assignment-symmetric equilibrium is positive exactly when

$$\boxed{xg'(x) - g(x) > a.} \quad (4)$$

At equality the checkerboard mode is marginal, and below it that mode is linearly stable. Hence affine nonnegative reinforcement, including $g(x) = bx$, cannot generate checkerboard instability. More generally, if $g(0) = 0$ and g is differentiable and concave on the physically accessible interval, then $xg'(x) \leq g(x)$, so (4) is impossible throughout that interval.

This boundary result forces a sufficiently superlinear local mechanism or a qualitatively different architecture; no parameter tuning can produce checkerboard symmetry breaking from a linear self-template within this model family.

4.3 Cooperative homotypic relational templating

The construction below uses quadratic reinforcement,

$$g(x) = bx^2,$$

interpreted as a coarse-grained cooperative rule in which already formed complexes of a given physical topology accelerate formation of another complex with the same topology. The rule is uniform over all i, j and does not read a target map. Homotypic recognition and topology copying are assumptions of this mechanism, not consequences of assignment neutrality.

The mean-field equations are

$$\boxed{\dot{x}_{ij} = (a + bx_{ij}^2)r_i e_j - dx_{ij}.} \quad (5)$$

The symmetric assignment mode becomes marginal at

$$x_c = \sqrt{a/b},$$

which is physically accessible only for

$$\boxed{b > 4a.} \quad (6)$$

The corresponding threshold is

$$\boxed{d_c = 2\sqrt{ab} \left(1 - 2\sqrt{a/b}\right)^2.} \quad (7)$$

Restricting to the checkerboard subspace

$$X = \begin{pmatrix} p & q \\ q & p \end{pmatrix}, \quad u = p + q, \quad m = p - q,$$

gives

$$\dot{u} = \left[2a + \frac{b}{2}(u^2 + m^2)\right] (1 - u)^2 - du, \quad (8)$$

$$\dot{m} = m[bu(1 - u)^2 - d]. \quad (9)$$

A nonzero branch obeys

$$\boxed{d = bu(1 - u)^2, \quad m^2 = u^2 - \frac{4a}{b}.} \quad (10)$$

The two states $m > 0$ and $m < 0$ are exactly related by output relabeling.

Theorem 4 — stable symmetry-broken coding branches

The broken branches are linearly asymptotically stable in the full four-dimensional relation system exactly when

$$\boxed{u > \frac{1}{3}.} \quad (11)$$

For $4a < b < 36a$, nonzero broken states sufficiently close to the continuous bifurcation lie on the stable side. For $b > 36a$, the pitchfork is subcritical; the stable broken segment $u > 1/3$ meets the unstable segment at the saddle-node $u = 1/3$. At that endpoint the linearization has a zero eigenvalue, so strict linear asymptotic stability is asserted only for $u > 1/3$. The saddle-node value is

$$\boxed{d_{SN} = \frac{4b}{27}.}$$

The coincident case $b = 36a$ is degenerate and is not included in either generic bifurcation description. Under the controlled readout assay of Section 2.3,

$$K(j \mid i; X) = \frac{x_{ij}}{x_{i0} + x_{i1}},$$

and on the broken branches

$$\boxed{C(K) = \frac{|m|}{u} = \sqrt{1 - \frac{4a}{bu^2}} > 0.} \quad (12)$$

Thus the symmetry-broken physical order is directly an operational, target-map-free coding order.

4.4 Finite-system branch choice and order of limits

For finite N , let n_{ij} be relation counts, with propensities

$$q_{ij}^{(a)} = a \frac{R_i^{free} E_j^{free}}{N}, \quad q_{ij}^{(b)} = b \frac{n_{ij}(n_{ij} - 1) R_i^{free} E_j^{free}}{N^3}, \quad q_{ij}^{(d)} = d n_{ij}. \quad (13)$$

The generator commutes with output swap ρ , and the empty initial state is fixed by ρ . Write the physical density space and its output-symmetric subset as

$$\mathcal{X}_{rel} = \{x_{ij} \geq 0 : \sum_j x_{ij} \leq 1, \sum_i x_{ij} \leq 1\},$$

$$S_\rho = \text{Fix}(\rho) \cap \mathcal{X}_{rel}.$$

Before choosing N , fix relatively open mature coding neighborhoods B_+ and $B_- = \rho B_+$ of the two stable broken equilibria. Require their relative closures in $\mathcal{X}_{rel}$ to be disjoint, to lie within their respective local attraction basins with positive readout row totals, and to satisfy

$$\delta_B = \text{dist}_\infty(\overline{B_+} \cup \overline{B_-}, S_\rho) > 0.$$

These requirements are realizable by sufficiently small swap-related balls: the two stable broken equilibria lie outside the closed set S_ρ , in the physical interior, and have positive readout row totals. Such fixed

balls contain feasible density lattice states for all sufficiently large N . Let $\tau_{\pm}$ be the hitting times of these neighborhoods by the density process and $\tau_{sel}^{(N)} = \min\{\tau_+, \tau_-\}$. The neighborhoods and their positive separation do not vary with N .

Theorem 5 — exact unbiased first-basin selection

For every finite N for which both basin neighborhoods contain feasible states,

$$\Pr(\tau_+ < \tau_-) = \Pr(\tau_- < \tau_+) = \frac{1}{2}. \quad (14)$$

Proof. Path-law invariance under ρ exchanges the two first-hit events. Irreducibility implies one of the nonempty basins is eventually hit almost surely, and disjointness excludes a tie. $\square$

This is the precise sense in which the coding organization is selected without an external mapping bias.

Theorem 6 — the fixed-time fluid limit does not select a branch

Starting from exactly symmetric data, the deterministic density trajectory remains in S_{ρ} . Fix a finite $T > 0$. Its distance from $B_+ \cup B_-$ is at least the prescribed $\delta_B > 0$ throughout $[0, T]$. Choose a compact ambient stopping region K containing the deterministic path with positive buffer

$$\delta_K = \inf_{0 \leq t \leq T} \text{dist}_{\infty}(\phi_t(X^N(0)), K^c) > 0, \quad r_T = \frac{1}{2} \min\{\delta_B, \delta_K\}.$$

For the empty initial state, this path, region and buffer are independent of N . More generally, the same argument applies to exactly symmetric initial data with a common compact buffered tube. Write $\tau_K^N = \inf\{t \geq 0 : X^N(t) \notin K\}$. Apply the stopped martingale concentration estimate in Appendix F to the fixed-volume specialization $g = 0$, so $v(t) \equiv 1$. On the physical part of the compact stopping region, the bounded-jump, bounded-rate and local-Lipschitz hypotheses hold, and the falling-factorial drift error is $O(1/N)$. For suitable constants $C_T, c_T > 0$, independent of N , and all sufficiently large N ,

$$\Pr\left(\sup_{0 \leq t \leq T \wedge \tau_K^N} \|X^N(t) - \phi_t(X^N(0))\|_{\infty} > r_T\right) \leq C_T e^{-c_T N}.$$

If a coding neighborhood is hit before leaving K , the trajectory deviation is at least $\delta_B > r_T$. If K is left first, the stopped deviation at that first exit is at least $\delta_K > r_T$. Thus the first-hit event by time T is contained in the stopped deviation event, and

$$\Pr(\tau_{sel}^{(N)} \leq T) \leq C_T e^{-c_T N} \rightarrow 0 \quad (N \rightarrow \infty). \quad (15)$$

The conclusion concerns these fixed, positively separated mature neighborhoods. Disjoint open neighborhoods alone would not provide the required positive distance.

Hence “spontaneous” in this paper refers to finite-system symmetry-respecting stochastic first-basin selection plus long residence. It does not mean that an exactly symmetric deterministic trajectory breaks symmetry at fixed time after taking $N \rightarrow \infty$ first.

For initial conditions in a compact inner neighborhood of a stable broken equilibrium, the same stopped estimate gives, for every fixed $L > 0$, $\Pr(\tau_{exit}^{(N)} \leq L) \rightarrow 0$. Therefore

$$\tau_{exit}^{(N)} \xrightarrow{P} \infty, \quad \mathbb{E}\tau_{exit}^{(N)} \rightarrow \infty. \quad (16)$$

The expectation statement follows because for every fixed L , $\mathbb{E}\tau_{exit}^{(N)} \geq L \Pr(\tau_{exit}^{(N)} > L)$, and the probability tends to one. No external density-dependent Markov convergence theorem is used in this proof.

Figure 2 summarizes the analytic instability, the two symmetry-related coding branches, and the finite-system/fluid-limit distinction.

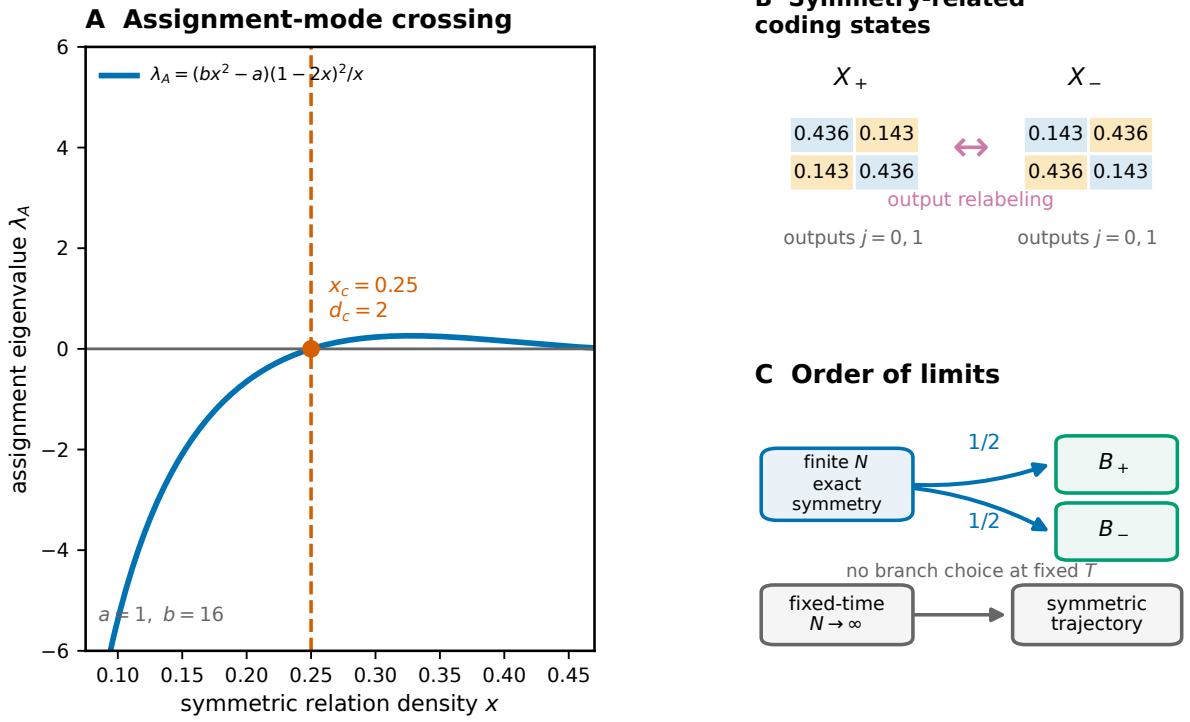


Figure 2: **Symmetry breaking without a target map.** **A**, for the quadratic CHRT family the checkerboard assignment eigenvalue crosses zero at the analytic threshold $x_c = \sqrt{a/b}$; the displayed representative case uses $a = 1, b = 16$. **B**, representative symmetry-related checkerboard equilibria X_+ and X_- are exchanged by output relabeling. **C**, exact finite- N path-law symmetry gives first-basin probabilities $1/2$ and $1/2$, while the exactly symmetric fixed-time fluid limit does not select a branch.

5. Reflexive maintenance with no stage switch

The coding scaffold still lacks reflexivity: a relation can mediate output without its outputs contributing to future interpreter capacity. We add a symmetry-blind support loop consistent with the stated causal constraints.

Let A_{ij} denote active relation complexes, D_{ij} damaged but topology-preserving relation complexes, and y a common maintenance currency. Define

$$r_i = 1 - \sum_j (A_{ij} + D_{ij}), \quad e_j = 1 - \sum_i (A_{ij} + D_{ij}).$$

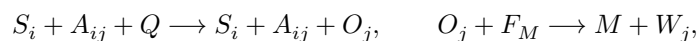
All rules are present from time zero. At the effective level,

$$\dot{A}_{ij} = (\alpha + bA_{ij}^2)r_i e_j - (d + \delta)A_{ij} + \rho y D_{ij}, \quad (17)$$

$$\dot{D}_{ij} = \delta A_{ij} - (\ell + \rho y)D_{ij}, \quad (18)$$

$$\dot{y} = \beta \sum_{ij} A_{ij} - \mu y - \rho y \sum_{ij} D_{ij}. \quad (19)$$

The maintenance source is shorthand for an explicit label-blind path



followed by topology-preserving repair

$$D_{ij} + M \longrightarrow A_{ij} + W_M.$$

Maintenance cannot create the first relation topology: D_{ij} arises only by damage of an already formed A_{ij} .

5.1 Symmetry-broken reflexive equilibrium

Take

$$A_+ = \begin{pmatrix} p & q \\ q & p \end{pmatrix}, \quad p > q > 0.$$

At equilibrium,

$$D_+ = cA_+, \quad c = \frac{\delta}{\ell + \rho y}.$$

Writing $u = p + q$ and $s = 1 - (1 + c)u$, the broken-state equations imply

$$\boxed{pq = \frac{\alpha}{b}.} \tag{20}$$

Thus the maintenance feedback does not define the assignment. The remaining equilibrium conditions are

$$bu \left[1 - \left(1 + \frac{\delta}{\ell + \rho y} \right) u \right]^2 = d + \frac{\delta \ell}{\ell + \rho y}, \tag{21}$$

$$2\beta u = \mu y + \frac{2\rho y \delta u}{\ell + \rho y}. \tag{22}$$

Theorem 7 — open region of stable reflexive coding attractors

Choose a hyperbolically stable broken equilibrium of the coding system at $\delta = 0$, with

$$u_0 > \max \left\{ \frac{1}{3}, 2\sqrt{\alpha/b} \right\}, \quad d = bu_0(1 - u_0)^2,$$

and arbitrary positive β, ρ, ℓ, μ . Then there exists $\varepsilon > 0$ such that every sufficiently small $0 < \delta < \varepsilon$ has two output-swap-related hyperbolically stable equilibria with

$$\boxed{C = \sqrt{1 - \frac{4\alpha}{bu^2}} > 0, \quad D > 0, \quad y > 0.} \tag{23}$$

Proof sketch. At $\delta = 0$, $y_0 = 2\beta u_0/\mu$. The reduced equations $F_1(u, y, \delta) = F_2(u, y, \delta) = 0$ have Jacobian determinant

$$-\mu b(1 - u_0)(1 - 3u_0) \neq 0.$$

The implicit-function theorem gives smooth $u(\delta), y(\delta)$. The strict coding inequality persists by continuity. At $\delta = 0$ the full Jacobian is block triangular with the stable coding block, four eigenvalues $-(\ell + \rho y_0)$, and one eigenvalue $-\mu$; hyperbolicity therefore persists for small positive δ . $\square$

Hyperbolicity and the strict inequalities imply a nonempty open symmetry-respecting parameter region, rather than an isolated fine-tuned point.

Theorem 8 — code-mediated outputs causally maintain the interpreter

At either reflexive coding equilibrium, perform a matched support block at $t = 0$: preserve the instantaneous state and matched burden, but prevent code-mediated output from producing new maintenance currency. The intact system remains at equilibrium. In the blocked system,

$$\dot{y}_B(0) = -\beta A_{tot}^*, \quad \dot{A}_B(0) = \dot{D}_B(0) = 0,$$

and therefore

$$\ddot{A}_{ij}^B(0) = -\rho\beta A_{tot}^* D_{ij}^*.$$

Hence

$$A_{ij}^{intact}(t) - A_{ij}^{block}(t) = \frac{\rho\beta A_{tot}^* D_{ij}^*}{2} t^2 + O(t^3). \quad (24)$$

Fix a common readout scale $k > 0$ and define the intervention-defined interpreter capacity by

$$\text{Cap}_I(A) = \frac{k}{2} \sum_j |A_{0j} - A_{1j}|.$$

At the chosen equilibrium, $D_* = c_* A_*$, so (24) gives

$$\Delta \text{Cap}_I(t) = \frac{\rho\beta A_{tot}^* c_*}{2} \text{Cap}_I(A_*) t^2 + O(t^3) > 0. \quad (25)$$

The causal loop

interpreter $\rightarrow$ decoded output $\rightarrow$ future interpreter capacity

is therefore closed at the maintenance level, with no rule switch triggered by detecting a code.

Corollary 2 — direct local consequence of disabling the readout path

A stricter C3 intervention disables transmission through the realized code/readout path itself while redirecting would-be readout events to an isomass/isoresource inert sink. In the reduced system, the code-mediated maintenance-production flux changes from

$$J_M^{code} = \beta A_{tot}^* > 0$$

to zero. Consequently

$$\int_0^t (J_{M,intact}^{code} - J_{M,disabled}^{code}) ds = \beta A_{tot}^* t + O(t^2) > 0 \quad (26)$$

for sufficiently small $t > 0$. This is the local causal consequence required for the main theorem; it is generated by explicit chemistry rather than a code-quality fitness score.

For the finite- η elementary CRN, the two intervention arms are defined without treating the pre-existing $O_j = O(\eta)$ stock as newly decoded output. At $t = 0$, **both** arms replace every current active O_j molecule by an equal-moiety, equal-load inert isomer O_j^0 . Thereafter the intact arm restores reaction (32), whereas the disabled arm redirects every would-be readout product to O_j^0 ; all other rates, challenges, boundary activities, and observation times are identical. This is an assay operation, not an added autonomous reaction or state variable in the unperturbed 115-channel process. If

$$r_O^* = \sum_{ij} \frac{k_Q Q^*}{\eta} U_{ij}^* > 0,$$

then the active-output and maintenance-production contrasts satisfy

$$J_{O,intact}^{new}(0) - J_{O,disabled}^{new}(0) = r_O^*,$$

$$\int_0^t (J_{M,intact}^{new} - J_{M,disabled}^{new}) ds = \frac{k_M F_M}{2\eta} r_O^* t^2 + O(t^3) > 0 \quad (26a)$$

for sufficiently small $t > 0$. Thus the exact elementary intervention has a strict finite- η consequence even though the inherited active-output stock is matched and removed from both arms.

5.2 Output-mediated maintenance and additional closure properties

Theorem 8 and Corollary 2 establish output-mediated maintenance under the stated assumptions. The following comparison separates that result from additional properties:

Property	Causal requirement	Status in this model
Output-mediated maintenance	Readout products contribute to a common process that preserves future interpreter capacity	Established under the stated assumptions
Mapping-dependent maintenance	Interpreter maintenance depends on which symbol–output assignment is realized	Not established
Encoded interpreter construction	Encoded products specify or build the machinery that interprets the same descriptions	Not established

Both O_0 and O_1 can contribute to the common maintenance currency M ; swapping the two coding orientations therefore need not reduce maintenance. The proved causal statement is that removing code-mediated output while matching immediate burden strictly reduces future interpreter capacity. This establishes a dependence on readout products beyond mere persistence of the channel. It does not establish that encoded products specify or build the interpreter.

These distinctions locate the result relative to physical-interpreter and semantic-closure accounts [Sharov 2009b; Wills 2009; Clark, Hickinbotham and Stepney 2017]. They are not intended as a general hierarchy of coding systems.

6. Elementary open reaction-network realization

The effective quadratic relation-templating term in Section 4 is useful for the bifurcation analysis, but it cannot be interpreted as an elementary collision. We therefore realize these slow dynamics in the singular limit of a low-molecularity mass-action network with explicit material and fuel bookkeeping.

6.1 Pair indices are compositional provenance, not a decoder table

For each receptor type R_i and effector type E_j , names such as A_{ij} , B_{ij} , or U_{ij} denote **assembled carriers containing both typed ports**. None is a primitive species supplied at the boundary. The first joint-index state, B_{ij} , is produced only by the uniform assembly schema $R_i + E_j \rightleftharpoons B_{ij}$. Every subsequent reaction that retains both indices has a pair-bearing carrier among its reactants. In particular, $A_{ij} + S_i \rightleftharpoons U_{ij}$ tests only the receptor-side port i , and $U_{ij} + Q \rightarrow A_{ij} + S_i + O_j$ actuates only the effector-side port j . The same kinetic form applies to every port value; there is no direct $S_i \rightarrow O_j$ rule and no branch or rate chosen by the ordered pair (i, j) .

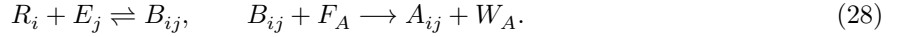
For the final growing robust process, use $C_R^* = C_E^* = 1$ and, for every integer $N \geq 1$, define the deterministic STRICT-NC count state z_0^N by

$$V_N(0) = N, \quad n_{R_0} = n_{R_1} = n_{E_0} = n_{E_1} = N, \quad (27)$$

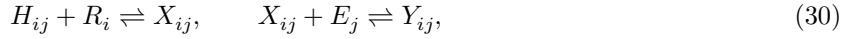
with all pair-bearing host coordinates B, A, D, U, H, X, Y , all parasite/error coordinates P, Z, X^P, Y^P , both outputs, and maintenance equal to zero. Set $\mu_0^N = \delta_{z_0^N}$. Symbol, substrate, and fuel activities are external, positive, and type-symmetric; wastes are external bookkeeping coordinates. When growth is active the first fission is scheduled at $T_g = \ln 2/g$. The host-only fixed-volume assay uses the corresponding 35-coordinate restriction and omits growth imports and parasite/error coordinates. Thus every positive integer N is admissible under the chosen normalization, and a pair-bearing state can appear only through an explicit construction history.

6.2 Elementary reaction grammar

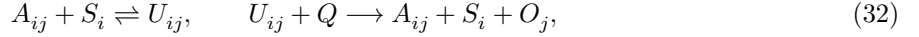
Basal relation nucleation is implemented by



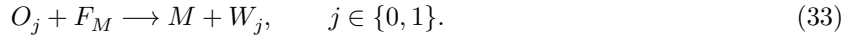
The cooperative relation-templating cycle is



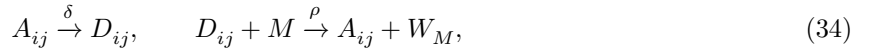
The symbol-conditioned readout is



and decoded output is converted into a common maintenance currency through



Finally,



with ordinary active/damaged turnover and maintenance decay. Every reaction family uses the same kinetic law for every symmetry-related pair.

Lemma 2 — low-molecularity admissibility

The explicit reaction inventory contains 71 instantiated reactions and every reactant side has molecularity at most two.

The reaction names in (28)–(34) suppress fast-scale coefficients for readability; Appendix E gives the scaled version used in the singular-perturbation proof.

6.3 Formal stoichiometry and energy boundary

Assign independent formal receptor moieties r_i , effector moieties e_j , symbol moieties s_i , read-out/maintenance moiety q , and fuel moieties f_A, f_R, f_M . The complete internal composition assignment is

$$[R_i] = r_i, \quad [E_j] = e_j, \quad [S_i] = s_i, \quad [A_{ij}] = [B_{ij}] = [D_{ij}] = [P_{ij}] = [Z_{ij}] = r_i + e_j,$$

$$[H_{ij}] = 2r_i + 2e_j, \quad [X_{ij}] = 3r_i + 2e_j, \quad [Y_{ij}] = 3r_i + 3e_j,$$

$$[U_{ij}] = r_i + e_j + s_i,$$

$$[X_{ij}^P] = 2r_i + e_j, \quad [Y_{ij}^P] = 2r_i + 2e_j,$$

$$[Q] = [O_j] = [M] = [W_M] = [W_M^{(0)}] = q,$$

$$[F_A] = [W_A] = f_A, \quad [F_R] = [W_R] = f_R, \quad [F_M] = [W_j] = f_M.$$

The import precursor R_i^{in} carries r_i , and E_j^{in} carries e_j . With fuels, substrates, symbols, and wastes included as boundary/bookkeeping species, every one of the 115 reaction instances conserves each declared formal moiety; only the four explicit material-import channels change the internal receptor/effector totals.

The driven directions are not assumed to be free. They require an open environment maintaining positive chemical affinities for relation activation, cooperative relation production, output-to-maintenance transduction, and repair. In particular, sustained support flux is interpreted as an open driven state.

This is a formal mass-action and stoichiometric embedding with irreversible driven reactions. No finite-affinity microscopic realization satisfying local detailed balance is constructed here. Equal formal moiety totals in the ideal interventions are bookkeeping constraints; they do not establish equal free energy or intervention work.

Theorem 9 — finite closed fuel excludes sustained positive support flux

Suppose a closed deterministic vessel begins with energetic inventory $E(0) < \infty$, no energetically equivalent regeneration, and each unit of driven support flux consumes at least $\varepsilon_E > 0$ units of that inventory. Then

$$\int_0^\infty J_{support}(t) dt \leq \frac{E(0)}{\varepsilon_E} < \infty.$$

Consequently the system admits no positive steady support flux, no $\liminf_{t \rightarrow \infty} J_{support}(t) > 0$, and no positive long-run time-average support flux. This does **not** exclude a transient flux that remains strictly positive at every finite time while decaying integrably.

For the stochastic CRN, if the energetic inventory is integer-valued, every support firing decreases it by at least one unit, and the process is nonexplosive, then the number of support firings is pathwise bounded by the initial inventory and is therefore finite almost surely. This theorem is an energetic boundary on the model. It does not imply that a particular prebiotic environment supplied the required fuel.

6.4 Singular-limit recovery of the effective coding dynamics

Introduce a small fast-intermediate occupancy/timescale parameter $\eta > 0$. On the attracting fast manifold, the constructed intermediates obey, uniformly on compact positive-resource sets,

$$B_{ij} = \eta K_B R_i E_j + O(\eta^2), \quad H_{ij} = \eta K_H A_{ij}^2 + O(\eta^2), \quad (35)$$

$$X_{ij} = \eta K_R K_H A_{ij}^2 R_i + O(\eta^2),$$

$$Y_{ij} = \eta K_E K_R K_H A_{ij}^2 R_i E_j + O(\eta^2), \quad (36)$$

$$U_{ij} = \eta K_S A_{ij} S_i + O(\eta^2). \quad (37)$$

Lemma 3 — active-unit slow-coordinate lifting for the elementary host

For each physical pair define the fast-reaction-invariant active-unit coordinate

$$a_{ij} = A_{ij} + U_{ij} + 2(H_{ij} + X_{ij} + Y_{ij}).$$

The fast cycles $2A \rightleftharpoons H$, $H + R \rightleftharpoons X$, $X + E \rightleftharpoons Y$, and $A + S \rightleftharpoons U \rightleftharpoons A + S + O$ cancel identically in $\dot{a}_{ij}$. For the scaled host grammar before compartment growth is added,

$$\dot{a}_{ij} = \frac{k_A F_A}{\eta} B_{ij} + \frac{k_{cat} F_R}{\eta} Y_{ij} - (d + \delta) A_{ij} + \rho M D_{ij},$$

$$\dot{D}_{ij} = \delta A_{ij} - (\ell + \rho M) D_{ij}.$$

Thus the potentially singular internal binding fluxes do not appear in the slow active-unit derivative. On the attracting fast manifold,

$$\begin{aligned} A_{ij} &= a_{ij} + O(\eta), \\ \frac{B_{ij}}{\eta} &= K_B R_i E_j + O(\eta), \\ \frac{Y_{ij}}{\eta} &= K_E K_R K_H a_{ij}^2 R_i E_j + O(\eta). \end{aligned}$$

For fixed receptor/effector totals, the exact host material bookkeeping gives

$$R_i = C_{R_i} - \sum_j (a_{ij} + D_{ij}) + O(\eta), \quad E_j = C_{E_j} - \sum_i (a_{ij} + D_{ij}) + O(\eta),$$

because the residual differences are sums of fast intermediates $B, X, Y = O(\eta)$. Consequently, for an assignment-neutral symbol environment $S_0 = S_1$, the elementary host slow field in coordinates (a, D, M) is the reduced field plus $O(\eta)$:

$$\dot{a}_{ij} = [\alpha_{eff} + b_{eff} a_{ij}^2] R_i E_j - (d + \delta) a_{ij} + \rho M D_{ij} + O(\eta), \quad (38)$$

with the analogous D - and maintenance equations differing from the reduced system by $O(\eta)$. When neutral growth is introduced later, the common dilution terms $-ga_{ij}$, $-gD_{ij}$, and $-gM$ are added, while the explicit material boundary controls the total-moiety coordinates separately.

Here

$$\alpha_{eff} = k_A F_A K_B, \quad b_{eff} = k_{cat} F_R K_H K_R K_E, \quad (39)$$

and, for the symmetric positive challenge activity $S_0 = S_1 = S_* > 0$ and readout-substrate activity $Q_* > 0$,

$$\beta_{eff} = k_Q Q_* K_S S_*. \quad (39a)$$

The output-to-maintenance rate $k_M F_M > 0$ controls the attracting output timescale and cancels from the leading slow production flux. Therefore the elementary parameters are matched to any selected positive reduced triple (α, b, β) as follows. Choose positive equilibrium constants K_B, K_H, K_R, K_E, K_S , positive reverse-rate scales, and positive boundary activities F_A, F_R, Q_*, S_* independently; choose the corresponding forward rates to realize those equilibrium constants; and define

$$k_A = \frac{\alpha}{F_A K_B}, \quad k_{cat} = \frac{b}{F_R K_H K_R K_E}, \quad k_Q = \frac{\beta}{Q_* K_S S_*}. \quad (39b)$$

All three derived rates are strictly positive, and the induced $\eta = 0$ slow field then has exactly $(\alpha_{eff}, b_{eff}, \beta_{eff}) = (\alpha, b, \beta)$.

Theorem 10 — preservation of reflexive coding by the elementary CRN

Let E_+ and E_- be hyperbolic asymptotically stable reflexive coding equilibria of a selected reduced system, with $C(K) > 0$, and construct a strictly positive microscopic rate/activity vector λ_* by (39b). For the concrete elementary grammar (28)–(34), Appendix E proves that both nested fast critical manifolds are smooth, normally hyperbolic and attracting; normal hyperbolicity is therefore a derived property of the reaction grammar rather than an independent assumption. By the successive attracting-layer argument detailed in Appendix E [Tikhonov 1952, pp. 576–577, Theorem 2; Fenichel 1979], there exists $\eta_0 > 0$ such that for every

$$0 < \eta < \eta_0$$

the elementary network has corresponding equilibria

$$E_{\pm}^{(\eta)} = E_{\pm} + O(\eta)$$

that remain hyperbolically asymptotically stable and satisfy

$$C(K_{\pm}^{(\eta)}) > 0.$$

Here $K_{\pm}^{(\eta)} = \mathcal{K}(E_{\pm}^{(\eta)})$ is the finite-window channel of Section 2.3. Since $L = A + U = a - 2(H + X + Y) = a + O(\eta)$ on the host branch and the limiting row totals are positive, the exact readout formula gives $C(K_{\pm}^{(\eta)}) = C(K_{\pm}^{(0)}) + O(\eta) > 0$ for every fixed $\tau > 0$.

The strict matched-output maintenance inequality in Theorem 8 and the exact finite- η readout-disable contrast (26a) also hold for sufficiently small η .

For the elementary support block, replace the product M in reaction (33) by the inert equal-moiety waste W_{M0} , retaining the reaction rate, fuel consumption, and every other reaction. This leaves the fast output clearance unchanged and removes only new output-mediated maintenance production. Measure future capacity in separate controlled-readout samples by

$$\text{Cap}_I^{\eta, \tau}(x) = \frac{\Gamma_{\eta, \tau}}{2} \sum_j |L_{0j} - L_{1j}|.$$

Choose a fixed sufficiently small positive observation time t at which the reduced contrast (25) is strict, and then choose η sufficiently small. The intact and blocked slow fields converge on that fixed time interval, while $L = a + O(\eta)$; hence the capacity contrast divided by the common positive factor $\Gamma_{\eta, \tau}$ converges to the strictly positive reduced contrast with $k = 1$. This proves the elementary strict inequality with this order of choices. The reduced t^2 expansion is not asserted uniformly down to $t = 0$ at fixed η .

Moreover, these conclusions are open in the **actual microscopic rate/activity vector**. There is a neighborhood W_λ of λ_* in the positive symmetry-respecting rate/activity space such that every $\lambda \in W_\lambda$ induces $(\alpha_{eff}, b_{eff}, \beta_{eff})$ close to (α, b, β) , retains both hyperbolic branches and all strict channel/maintenance contrasts, and has a possibly smaller positive $\eta_0(\lambda)$. Exact macro matching is used to construct the witness point λ_* ; nearby microscopic vectors need not satisfy the three equalities exactly.

At $g = 0$, the four receptor/effector total moieties are conserved. Accordingly, hyperbolicity of the fixed-volume elementary host is always understood **relative to the fixed stoichiometric compatibility class** (equivalently, after eliminating the four conserved-total directions); the ambient Jacobian has four conservation zero modes. When the explicit growth-material boundary of Section 7.3 is activated with $g > 0$, those four total-moiety directions become the exact transverse modes $-g$.

Scope. This establishes a chemically interpretable mass-action embedding at the declared abstraction level. It does not establish the prebiotic availability of the molecular species, fuels, or reaction families.

Theorem 10A — fixed-volume elementary stochastic genesis

Fix $0 < \eta < \eta_0$, a finite volume scale N , strictly positive chemostats/rates, and an output-swap-invariant integer conservation class with equal output-side conserved moiety totals. Treat chemostats as boundary parameters and accumulating wastes as external sinks/bookkeeping coordinates rather than recurrence-state coordinates. On the resulting projected/open finite-volume elementary CTMC, every receptor/effector-bearing structural coordinate is bounded by the four conserved receptor/effector moiety totals, while the only unbounded internal coordinates (O_0, O_1, M) have linear clearance. With

$$\Phi = 2(O_0 + O_1) + M,$$

the generator satisfies a Foster–Lyapunov drift inequality $\mathcal{L}\Phi \leq C_N - \epsilon\Phi$ for some finite C_N and $\epsilon > 0$ [Meyn and Tweedie 1993]. The reaction grammar also provides positive-rate clearance paths from every internal structural state to the STRICT-NC reference state. For the coding-basin lattice representatives used below, explicit forward reaction paths construct those target states from STRICT-NC. More generally, reachability is asserted only for states belonging to the declared communicating class; compatibility with the four receptor/effector moiety totals alone is **not** claimed to imply reachability. Hence the declared communicating class is nonexplosive, irreducible, and positive recurrent.

Therefore every nonempty feasible coding-basin lattice set is hit almost surely. If $B_-^{N,\eta} = \rho B_+^{N,\eta}$, the two sets are disjoint, and the STRICT-NC reference state is fixed by output swap ρ , then

$$\Pr(\tau_+ < \tau_-) = \Pr(\tau_- < \tau_+) = \frac{1}{2}.$$

This almost-sure statement is deliberately restricted to the fixed-volume elementary genesis CTMC. The growing/fission lineage below uses only positive-probability genesis and, when first-hit symmetry is invoked, a single-lineage output-swap-equivariant conditional statement; fixed-volume recurrence is not transferred to the time-inhomogeneous lineage process.

Proposition 1 — positive-support genesis in the full growing process

Fix the full microscopic parameter vector, including finite $\eta > 0$, $g > 0$, and either $\nu = 0$ or $\nu > 0$ in the robust range. Start the 115-channel time-inhomogeneous process from μ_0^N in (27), and let $T_G < T_g$. For every sufficiently large admissible N , choose disjoint output-swap-related lattice neighborhoods B_+^N and $B_-^N = \rho B_+^N$ contained in the two robust coding basins. Then T_G and the neighborhoods can be chosen so that

$$\Pr_{\mu_0^N}(\tau_+ < \tau_-, \tau_+ \leq T_G) = \Pr_{\mu_0^N}(\tau_- < \tau_+, \tau_- \leq T_G) > 0. \quad (39c)$$

Consequently, conditional on hitting their union by T_G , the first orientation is exactly unbiased.

Proof. Before the first fission, the symmetric import channels can supply any finite additional stock of free R_i, E_j . From those stocks, a finite ordered sequence of positive-rate reactions constructs basal carriers and active relation units of one checkerboard orientation. Further finite subsequences create the required

H, X, Y, U, O, M, D counts. When $\nu > 0$, error, parasite binding, damage, and repair reactions similarly create the required P, X^P, Y^P, Z counts; when $\nu = 0$, the target lattice state has those coordinates zero. Free receptor and effector residuals are set by the number of imports. Because the target basin is open and lattice spacings are $O(1/N)$, a reachable count vector lies inside it for all sufficiently large N .

Along this finite path every scheduled reaction has strictly positive propensity once its reactants have been prepared. On a compact time interval the probability that these reactions occur in the prescribed order, with their event times in prescribed nonempty subintervals and with no competing event before the target is reached, is a finite product of strictly positive exponential-race factors. Lemma 4 ensures the complete process is nonexplosive. Hence the path event has positive probability. The time-dependent generator, the deterministic volume protocol, μ_0^N , and the later single-daughter fission kernel all commute with output swap ρ ; applying ρ maps the positive path event bijectively to the opposite first-hit event and proves (39c). $\square$

This proposition proves only a positive-support route from the specified STRICT-NC state. It does not give a useful lower bound, a typical waiting time, or a high-probability route to the mature-parent class used in the heredity theorem.

ANALYSIS / ASSAY

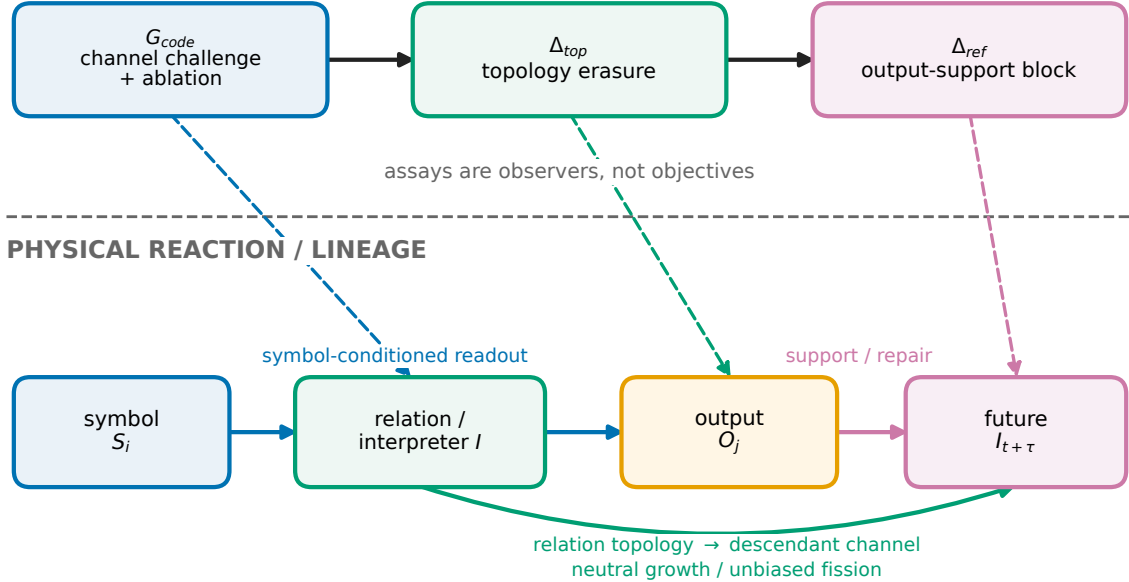


Figure 3: **Causal Closure Triangle.** The analysis layer contains three separately matched interventions: the operational code gate G_{code} , topology-heredity contrast Δ_{top} , and reflexive-support contrast Δ_{ref} . Dashed connectors show which physical edge each assay interrogates. The physical dynamics do not evaluate these analysis-side statistics as objectives.

7. Compartment lineage and causal heredity

Operational coding and reflexive maintenance do not by themselves establish heredity. Composition-based inheritance and its evolvability limits have been studied in dividing molecular assemblies [Segré, Ben-Eli and Lancet 2000; Vasas, Szathmáry and Santos 2010], while experimental compartmentalized autocatalytic networks provide a related physical comparator [Ameta et al. 2023]. The criterion used here is narrower and causal: parental relation topology must change descendant channel distributions under a matched topology-erasure intervention. We therefore add only a mapping-neutral compartment lineage process.

7.1 Unbiased molecule-level fission

Let $\mathcal{K}(x)$ denote the intervention channel obtained from physical state x by the fixed challenge protocol of Section 2.3. Every required challenge uses the same symbol activity, resource activities, assay-window length, and observation time. For the two robust orientations let K_+^* and K_-^* be their reference channels and define

$$\mathcal{N}_\sigma(\epsilon) = \{K : d_K(K, K_\sigma^*) \leq \epsilon\}, \quad \sigma \in \{+, -\},$$

with $0 < \epsilon < \frac{1}{2}d_K(K_+^*, K_-^*)$, so the two channel neighborhoods are disjoint.

We first isolate a **pure-partition assay**. Start with a mature parent compartment of volume $4N$. Every internal molecular object is independently assigned to either daughter with probability $1/2$, irrespective of species, pair topology, channel orientation, or any analysis-side statistic. Apply two fissions immediately, with no chemistry, recovery, import, or growth between them, and follow one specified daughter at each split:

$$4N \longrightarrow 2N \longrightarrow N. \quad (40)$$

If the parent contains n_s copies of internal species s , then the focal child and grandchild counts satisfy

$$Y_{s,1} \sim \text{Bin}(n_s, 1/2), \quad Y_{s,2} \sim \text{Bin}(n_s, 1/4). \quad (41)$$

If $n_s \leq 4NM_{P2}$ on a compact coding-basin core, Hoeffding's inequality [Hoeffding 1963] gives

$$\Pr(|X_{s,1} - x_s| > r) \leq 2e^{-2Nr^2/M_{P2}}, \quad (42)$$

$$\Pr(|X_{s,2} - x_s| > r) \leq 2e^{-Nr^2/(2M_{P2})}. \quad (43)$$

For the host-only state dimension $D_{\text{host}} = 35$,

$$\Pr\left[\max_{g=1,2} \|X_g - x\|_\infty > r\right] \leq 4D_{\text{host}}e^{-Nr^2/(2M_{P2})}. \quad (44)$$

Because the intervention-defined channel is continuous on a compact interior coding core, choose ϵ_H smaller than half the separation between the two symmetry-related channels. There is then an $r_H > 0$ such that the concentration event in (44) implies both child and grandchild remain within ϵ_H of the parent channel.

Explicit matched topology-erasure kernel. For each sufficiently large admissible N , choose a balanced mature parent lattice state approximating either checkerboard equilibrium and lying in the compact coding core, with the declared receptor/effector-side marginals and other matched nuisance coordinates exactly invariant under output swap. For every host pair-bearing structural class $c \in \mathcal{C}_{\text{pair}}^{\text{host}} := \{B, A, D, U, H, X, Y\}$, keep the receptor-side and effector-side class marginals fixed and uniformly randomize the bijection between receptor-side and effector-side object slots. Equivalently, the induced 2×2 contingency table has the corresponding hypergeometric rewiring law. Apply these rewiring kernels independently across structural classes and leave all declared non-pair matched coordinates unchanged; call the product Markov kernel $\mathcal{E}$.

The required volume- $4N$ balanced lattice parents exist for all sufficiently large N . For a checkerboard target concentration table $[[p_c, q_c], [q_c, p_c]]$, choose

$$P_{c,4N} = \text{round}(4Np_c), \quad Q_{c,4N} = \text{round}(4Nq_c),$$

and use the exact count table $[[P_{c,4N}, Q_{c,4N}], [Q_{c,4N}, P_{c,4N}]]$. Round output-indexed nuisance coordinates in equal pairs and define free R_i, E_j counts as residuals of the exact volume- $4N$ target moiety totals. The concentration error is $O(1/N)$, retained row/column symmetries are exact, and positive limiting free material makes every residual nonnegative for sufficiently large N .

Uniform slot matching is equivariant: $\mathcal{E}(\rho x, \rho A) = \mathcal{E}(x, A)$. Because the retained statistic of the balanced parent is itself output-swap invariant, the erased-parent law $\mathcal{E}(x, \cdot)$ is exactly ρ -invariant. The two pure partition kernels commute with the same swap. Therefore the erased descendant law is ρ -invariant, and each of two disjoint swapped orientation events has probability at most $1/2$.

Theorem 11 — two-fission pure-partition heredity assay

For a balanced volume- $4N$ mature parent lattice state x in orientation σ , fix $K_{parent} := \mathcal{K}(x)$ from the intact parent before any intervention. Use this same deterministic reference in both the intact and topology-erased experimental arms; it is not recomputed from the randomized erased parent. Choose $\epsilon_H > 0$ with a uniform strict separation margin on the selected compact mature-parent core, so in particular $2\epsilon_H < d_K(K_{parent}, \rho K_{parent})$; this is possible in a sufficiently small core around the broken equilibrium, and holds for its balanced lattice approximations for all sufficiently large N . Let $K_{child} = \mathcal{K}(X_1)$ and $K_{grandchild} = \mathcal{K}(X_2)$ be the channels associated with the two immediate pure-partition descendants. These are ratios of conditional expected new output counts in the volume- $2N$ and volume- N side samples, respectively; the exact formula of Section 2.3 gives the same $\mathcal{K}$ at either volume. On the compact coding core, choose a positive lower bound ℓ_0 on both row totals of L . Then normalization is Lipschitz there, so continuity of $\mathcal{K}$ gives $r_H > 0$ such that

$$\Pr [d_K(K_{child}, K_{parent}) \leq \epsilon_H, d_K(K_{grandchild}, K_{parent}) \leq \epsilon_H] \geq 1 - \eta_{P2}(N), \quad (45)$$

where

$$\eta_{P2}(N) = 4D_{host}e^{-Nr_H^2/(2M_{P2})}. \quad (46)$$

Let A_σ^{P2} be the event in (45), let $\mathbb{P}_\sigma^{P2,int}$ start from the intact parent, and let $\mathbb{P}_\sigma^{P2,erase}$ first apply $\mathcal{E}$ and then use the same two partition kernels and the same channel observations. The kernel preserves the declared receptor/effector marginals, structural-class totals, bulk material/resource load, and narrowly defined matched burden while randomizing receptor-effector partner correlations. The two closed channel neighborhoods of radius ϵ_H around K_{parent} and ρK_{parent} are disjoint. Their corresponding descendant events are swapped by ρ , while the erased law is invariant. Hence $\mathbb{P}_\sigma^{P2,erase}(A_\sigma^{P2}) \leq 1/2$. Therefore

$$\Delta_\sigma^{P2}(N) = \mathbb{P}_\sigma^{P2,int}(A_\sigma^{P2}) - \mathbb{P}_\sigma^{P2,erase}(A_\sigma^{P2}) \geq \frac{1}{2} - 4D_{host}e^{-Nr_H^2/(2M_{P2})} > 0 \quad (47)$$

for sufficiently large N . The transmitted trait is physical relation/interpreter topology and the induced channel, not a copied mapping label.

Theorem 11 is deliberately limited to two immediate partitions. It makes no claim about intervening chemistry or recovery. For the full lineage, define the **generation skeleton** X_h^N at the post-fission reset volume N . From X_h^N , the complete chemistry and symmetric imports run while volume grows from N to $2N$ over T_g , after which unbiased fission selects one daughter and resets its volume to N , producing X_{h+1}^N . The side-assay channel $K_h = \mathcal{K}(X_h^N)$ is evaluated at the same post-fission time with the same challenge and observation schedule in every generation and in both intervention arms.

For compact orientation-specific sets $D_{in}^\sigma \subseteq D_{out}^\sigma$ introduced in Section 7.4, define

$$\tau_{exit}^{N,\sigma} = \inf\{h \geq 0 : X_h^N \notin D_{out}^\sigma\}, \quad A_\sigma(H, \epsilon) = \{K_H \in \mathcal{N}_\sigma(\epsilon)\}.$$

The intact law $\mathbb{P}_\sigma^{int}$ starts from the specified balanced parent. The erased law $\mathbb{P}_\sigma^{erase}$ applies the stated topology-erasure kernel at generation zero and thereafter uses exactly the same chemistry, volume protocol, fission law, challenges, and observation times. Theorem 15, not Theorem 11, controls this full-cycle process.

7.2 Information cuts and the passive-segregation limit

A **parent-orientation information cut** is a physical carrier set whose complete loss reduces recovery of the named parent orientation to at most a declared baseline $b < 1$. Suppose the cut contains m ancestral copies. For copy k , let $I_{k,H}$ indicate that its parent-orientation information reaches the focal

descendant after H divisions. Assume that at every lineage edge, conditional on the complete prior history and on that copy being informative at the edge's start, the probability that it both remains informative and is delivered to the focal daughter is at most $\varrho/2$. This condition holds, for example, when chemical information retention is at most ϱ and the subsequent unbiased partition delivers the surviving carrier with conditional probability $1/2$. Iteration gives the marginal bound

$$\Pr(I_{k,H} = 1) \leq q_H := \left(\frac{\varrho}{2}\right)^H. \quad (48)$$

No independence between different ancestral copies, different retention events, or different lineage edges is assumed in this marginal statement.

Theorem 12 — dependence-robust and independent-copy cut ceilings

Let $Z_H = \sum_{k=1}^m I_{k,H}$. For the event $A_\sigma(H, \epsilon_H)$ that the focal descendant recovers the named parental channel within tolerance, the complete-loss definition and the union bound give, without an independence assumption,

$$\Pr(A_\sigma) \leq b + (1 - b) \min\{1, mq_H\}. \quad (49)$$

If, in addition, the m copy histories are mutually independent, every copy has exact survival probability q_H , and the partition and information-retention trials along its path are independent with the stated probabilities, then $Z_H \sim \text{Bin}(m, q_H)$ and the sharper ceiling is

$$\Pr(A_\sigma) \leq 1 - (1 - b)(1 - q_H)^m. \quad (49a)$$

Only under these additional independent-copy assumptions, a target reliability $1 - \delta > b$ requires

$$m \geq \left\lceil \frac{\ln[\delta/(1 - b)]}{\ln(1 - q_H)} \right\rceil. \quad (50)$$

For binary symmetry $b = 1/2$, two divisions, and $\varrho = 1$, the necessary copy counts are $m \geq 6, 9, 14$ for 90%, 95%, and 99% reliability, respectively. These are necessary, not sufficient, conditions.

For physically disjoint required cuts $a = 1, \dots, r$, put $B = \bigcup_a \{Z_{a,H} = 0\}$. The following product ceiling requires the **joint failure baseline** $\Pr(A_\sigma | B) \leq b$ whenever $\Pr(B) > 0$, in addition to independent exact copy survival within and across cuts. A sufficient stronger condition is a baseline at most b conditional on each complete cut-survival pattern with at least one failed cut. Under these hypotheses, conditioning on B and its complement gives

$$\Pr(A_\sigma) \leq b + (1 - b) \prod_{a=1}^r [1 - (1 - q_a)^{m_a}]. \quad (51)$$

Indeed, $\Pr(A_\sigma) \leq b \Pr(B) + \Pr(B^c)$, and independence factors $\Pr(B^c)$. Separate bounds $\Pr(A_\sigma | Z_{a,H} = 0) \leq b$ for individual cuts do not imply the joint failure baseline. For example, independent fair bits I_1, I_2 and $A = \{I_1 = 1 \text{ or } I_2 = 1\}$ give both separate baselines $1/2$, but $\Pr(A) = 3/4 > 5/8$, the unqualified product expression. Without the joint baseline or the stated independence, use (49) for any necessary single cut.

If a mapping-neutral information-loss probability μ acts independently on every lineage edge, $\varrho = 1 - \mu$. Under the same independent-copy hypotheses, the single-cut target reliability becomes impossible above

$$\mu_c = 1 - 2 \left[1 - \left(\frac{\delta}{1 - b} \right)^{1/m} \right]^{1/H}, \quad (52)$$

provided the zero-corruption partition ceiling can reach the target at all.

Theorem 13 — passive finite ancestral memory cannot support indefinite above-baseline heredity

For fixed finite m with no endogenous replenishment and any dependence structure satisfying the marginal bound above,

$$q_H \rightarrow 0 \quad (H \rightarrow \infty),$$

and the dependence-robust ceiling (49) gives

$$\boxed{\limsup_{H \rightarrow \infty} \Pr(A_\sigma) \leq b.} \quad (53)$$

For the binary symmetric case the limit superior is at most $1/2$. Long-horizon heredity therefore requires replenishment, active segregation, or another physical memory channel; passive segregation of a fixed finite ancestral stock is insufficient.

7.3 Neutral growth and endogenous replenishment

Let compartment volume obey

$$\dot{V} = gV \quad (54)$$

with a code-blind growth rate g . The open growth boundary is explicit rather than implicit. For each receptor and effector type introduce a chemostatted precursor carrying the same declared material moiety,

$$R_i^{\text{in}} \xrightarrow{g} R_i, \quad E_j^{\text{in}} \xrightarrow{g} E_j,$$

with equal precursor activities C_R^* for all receptor types and C_E^* for all effector types. In a compartment of volume $V_N(t)$, the corresponding count intensities are $gC_R^*V_N(t)$ and $gC_E^*V_N(t)$. These channels are domain-local, type-symmetric, code-blind, and vanish identically in the separately declared fixed-volume host assay $g = 0$. The 71-channel assay of Theorem 10A is host-only by definition; it is **not** obtained from the full 115-channel robust grammar merely by setting $g = 0$.

Let C_{R_i} and C_{E_j} denote total receptor/effector moiety concentrations, including the appropriate multiplicities in every host or parasite complex. Since every internal reaction conserves these moieties exactly, the **deterministic mass-action concentration equations** satisfy

$$\dot{C}_{R_i} = g(C_R^* - C_{R_i}), \quad \dot{C}_{E_j} = g(C_E^* - C_{E_j}).$$

For the normalized growing CTMC, the same expressions are the exact **compensator drifts** of the total-moiety concentration coordinates; individual sample paths additionally contain the import jump-martingale terms. No sample-path ODE identity is asserted.

Hence the target-total manifold $C_{R_i} = C_R^*$, $C_{E_j} = C_E^*$ is invariant and exponentially attracting with transverse eigenvalues $-g$. On that manifold the free concentrations are exactly target total concentration minus **all** bound material. Lemma 3 shows that, in the elementary host slow coordinates, these free-resource formulas and the internal vector field agree with the reduced growth model only up to $O(\eta)$; no finite- η exactness is asserted. We use the representative normalization $C_R^* = C_E^* = 1$.

For the **reduced $\eta = 0$ growth system**, the exact equilibrium relation

$$pq = \frac{\alpha}{b} \quad (55)$$

holds, and its reduced coding specificity is

$$C_g = \sqrt{1 - \frac{4\alpha}{bu_g^2}} > 0. \quad (56)$$

Both (55) and (56) are reduced/effective-model identities.

Theorem 14 — small neutral growth preserves the coding branch and replenishes carrier copy number

There exists $g_* > 0$ such that for every fixed $0 < g < g_*$ the reduced system has the two symmetry-related hyperbolically stable coding equilibria described by (55)–(56). For each such g , there exists $\eta_H(g) > 0$ such that, for

$$0 < \eta < \eta_H(g),$$

the elementary growing host has corresponding hyperbolically stable equilibria on the exact total-material manifold with

$$E_{H,\pm}^{(\eta,g)} = E_{\pm}^{(0,g)} + O(\eta), \quad C_H^{(\eta,g)} = C_g + O(\eta) > 0.$$

The specificity here is evaluated by $\mathcal{K}$ with the controlled readout protocol. The identity $L = a - 2(H + X + Y)$, the fast-carrier $O(\eta)$ bounds, and positive limiting row totals justify the stated channel error without assuming that a full-chemistry transient assay has the same response.

The exact growth-material balance equations control only the total receptor/effector moieties and supply four transverse modes $-g$; the internal finite- η slow dynamics are the reduced growth/dilution field plus $O(\eta)$, by Lemma 3 and nested normal-hyperbolic persistence. After the small label-blind error continuation of Section 8.2 the corresponding specificity obeys

$$C^{(\eta,g,\nu)} = C_g + O(\eta) + O(\nu) > 0$$

for sufficiently small η, ν . At balanced growth the volume-doubling time is

$$T_g = \frac{\ln 2}{g}. \tag{57}$$

Concentrations are stationary on the mature branch, so the absolute number of every internally produced pair-bearing carrier doubles during one growth cycle. Since pair-bearing complexes are not supplied through the boundary, this replenishment is generated by the internal templating/repair chemistry rather than by a mapping-copy instruction.

7.4 Long stochastic lineages and the finite-system reset boundary

Lemma 4 — finite-horizon nonexplosion of the final 115-channel growing process

Fix $N < \infty$ and a finite time horizon T . The four receptor/effector material imports are inhomogeneous Poisson channels with finite means

$$\int_0^T g C^* N e^{gs} ds = N C^* (e^{gT} - 1) < \infty.$$

Every internal host, parasite, and error reaction conserves the receptor/effector moieties. On a fixed finite growth interval, first stop the process when the total receptor/effector material count exceeds a deterministic level K . Before this stopping time, every structural count is bounded. Set $W = n_{O_0} + n_{O_1} + n_M$. Only readout increases W , and its rate is bounded by the structural ceiling. Conversion of output to maintenance preserves W , while repair and decay decrease it. Thus the upward jumps of W are dominated by a finite-rate immigration process. The complete reaction intensity is at most $C_{K,N,T}(1+W)$, including the maintenance-dependent repair reactions. Hence the stopped process cannot explode. The import counts are almost surely finite on the interval, so letting K increase proves nonexplosion of the full 115-channel time-inhomogeneous CTMC. The single scheduled fission at the cycle endpoint preserves this conclusion.

Lemma 5 — local one-cycle return geometry

Fix a continuous parameter vector θ for which the deterministic growing concentration field has a hyperbolically asymptotically stable mature coding equilibrium $x_*^\sigma(\theta)$, and fix the doubling time $T_g > 0$. Write $\mathcal{X} = \mathbb{R}_{\geq 0}^{51}$ for the physical concentration space. At zero error the equilibrium lies on its parasite-free boundary. Extend the polynomial deterministic field to an ambient open neighborhood of this point solely for the local flow argument; stochastic states and reaction intensities are always restricted to $\mathcal{X}$. Let Φ_t^θ denote that local flow.

There are compact physical neighborhoods $D_{in}^\sigma, D_{out}^\sigma$ and a compact **ambient** neighborhood K^σ such that

$$\begin{aligned} D_{in}^\sigma &\subset \text{int}_{\mathcal{X}} D_{out}^\sigma, \\ \Phi_{T_g}^\theta(D_{out}^\sigma) &\subset \text{int}_{\mathcal{X}} D_{in}^\sigma, \\ \{\Phi_t^\theta(x) : 0 \leq t \leq T_g, x \in D_{out}^\sigma\} &\subset \text{int } K^\sigma. \end{aligned}$$

Here $\text{int}_{\mathcal{X}}$ means interior in the topology of the physical orthant, not interior in its affine hull. The compact trajectory tube has positive distance from the ambient boundary ∂K^σ . Rate bounds on this stopped region are evaluated only on $K^\sigma \cap \mathcal{X}$. The neighborhoods lie in the same physical coding basin and satisfy $\mathcal{K}(D_{out}^\sigma) \subset \mathcal{N}_\sigma(\epsilon_H)$. Below, compact-containment notation between the D sets is understood in this relative physical topology; containment in K is ambient.

Proof. Let $J_* = DF_g(x_*^\sigma)$. Stability makes J_* Hurwitz, including the transverse parasite directions at zero error, so

$$A = D\Phi_{T_g}(x_*^\sigma) = e^{J_* T_g}$$

has spectral radius below one. The series

$$P = \sum_{n=0}^{\infty} (A^T)^n A^n$$

converges and satisfies $A^T P A = P - I$. In the norm $\|z\|_P^2 = z^T P z$,

$$\|Az\|_P^2 = \|z\|_P^2 - \|z\|_2^2 \leq q_0^2 \|z\|_P^2$$

for some $q_0 < 1$. Smoothness of the ambient flow gives a sufficiently small ball on which the nonlinear time- T_g map has Lipschitz constant $q \in (q_0, 1)$. Choose $qr_{out} < r_{in} < r_{out}$, and set

$$D_{in/out}^\sigma = \{x : \|x - x_*^\sigma\|_P \leq r_{in/out}\} \cap \mathcal{X}.$$

Positive invariance of $\mathcal{X}$ and the contraction imply the relative-interior return inclusion. The radius gaps give positive distances to the physical complements, even when x_*^σ has zero parasite coordinates. After a further shrink, the whole trajectory tube lies in the ambient basin and admits a compact ambient buffer K^σ with positive boundary distance. Continuity of the exact readout map, positive row totals, and separated reference channels allow a final shrink giving the channel inclusion. No physical open ball about the zero-error equilibrium is assumed. $\square$

For each fixed θ , including finite positive η and g , we use the neighborhoods supplied by Lemma 5. They are chosen after θ and before N, H ; no unproved common basin geometry over the entire compact parameter core is assumed. Let $\mathcal{F}_h$ be the complete history through the h -th post-fission observation. Every one-cycle probability bound below is uniform over all $\mathcal{F}_h$ -measurable states $X_h^N \in D_{out}^\sigma(\theta)$; this is the conditional uniformity needed to iterate the estimate.

During one maturation cycle let

$$v(t) = e^{gt}, \quad V_N(t) = Nv(t), \quad 0 \leq t \leq T_g = \frac{\ln 2}{g},$$

so $1 \leq v(t) \leq 2$. If $Z^N(t)$ is the count vector and $X^N(t) = Z^N(t)/(Nv(t))$, the exact stochastic mass-action intensity of reaction r can be written

$$\lambda_r^N(t, z) = Nv(t) a_r^{N, v(t)} \left(\frac{z}{Nv(t)} \right),$$

where the falling-factorial mass-action rate $a_r^{N, v}$ differs from its continuous density rate a_r by $O(1/N)$ uniformly on the stopped compact set because every internal reaction has molecularity at most two. The four growth-material import channels are included among the reaction channels with exact density rates gC_R^* and gC_E^* , so they contribute no falling-factorial bias. The concentration process therefore has drift $F_g(x) = F(x) - gx$, where F now includes the explicit material imports, an $O(1/N)$ finite-size bias from the internal molecular reactions, and martingale jumps $\nu_r/(Nv(t))$.

Let

$$J = \max_r \|\nu_r\|_\infty, \quad V_* = \max_i \sum_r \nu_{r,i}^2 \bar{a}_r, \quad \Xi_g(T) = \int_0^T e^{-gs} ds = \frac{1 - e^{-gT}}{g}.$$

Then every martingale jump is at most J/N , while

$$\langle M_i^N \rangle_T \leq \frac{V_* \Xi_g(T)}{N}.$$

Appendix F gives the resulting Freedman–Grönwall bound including the $O(1/N)$ stochastic-mass-action drift bias. If $r_P(\theta) > 0$ is the partition margin, $r_F(\theta) > 0$ the deterministic return margin, $r_K(\theta) > 0$ the trajectory-tube margin, and $M_*(\theta)$ bounds mature concentrations, set $\epsilon_F = \frac{1}{2} \min\{r_F, r_K\}$. Choose $N_0(\theta)$ large enough to absorb the finite-size drift bias.

The final robust count vector has

$$D_{state} = 7 \times 4 + 4 \times 4 + 2 + 2 + 2 + 1 = 51$$

internal coordinates: seven host pair tables, four parasite/error pair tables, two free-receptor and two free-effector coordinates, two outputs, and maintenance. A union bound contributes at most $2D_{state}$ terms for maturation and $2D_{state}$ for partition. Thus set

$$\begin{aligned} C_0 &= 4D_{state} = 204, \\ c(\theta) &= \min\{c_P(\theta), c_M(\theta)\} > 0, \\ c_P(\theta) &= \frac{r_P(\theta)^2}{4M_*(\theta)}, \\ c_M(\theta) &= \frac{z_F(\theta)^2}{2(V_*(\theta)\Xi_g(T_g) + Jz_F(\theta)/3)}, \\ z_F(\theta) &= \frac{\epsilon_F(\theta)e^{-L_g(\theta)T_g}}{2}. \end{aligned}$$

Consequently, for $N \geq N_0(\theta)$, the one-generation failure probability obeys

$$\epsilon_N \leq C_0 e^{-c(\theta)N}. \quad (58)$$

The order of limits is part of the statement: θ , $D_{in}^\sigma(\theta)$, $D_{out}^\sigma(\theta)$, and $K^\sigma(\theta)$ are fixed before N , and H is chosen last. No joint limit $\eta = \eta_N \rightarrow 0$ or $g = g_N \rightarrow 0$ is asserted.

Theorem 15 — exponentially long finite- N causal heredity

Start the intact full-generation skeleton from the exact-balanced post-fission parent $x_{N,\theta}^\sigma \in D_{in}^\sigma$, and define $\tau_{exit}^{N,\sigma}$ as in Section 7.1. Uniformly conditional on $\mathcal{F}_h$ while the process remains in D_{out}^σ , (58) bounds the next-cycle failure. Therefore, for H consecutive generations,

$$\boxed{\mathbb{P}_\sigma^{int}(\tau_{exit}^{N,\sigma} > H) \geq 1 - HC_0 e^{-cN}.} \quad (59)$$

Consequently, if

$$H_N \leq e^{\gamma N}, \quad 0 < \gamma < c(\theta),$$

then

$$\mathbb{P}_\sigma^{int}(\tau_{exit}^{N,\sigma} > H_N) \rightarrow 1. \quad (60)$$

Because $\mathcal{K}(D_{out}^\sigma) \subset \mathcal{N}_\sigma(\epsilon_H)$, survival implies $A_\sigma(H, \epsilon_H)$. Apply $\mathcal{E}_{full}$ at generation zero to the same exact-balanced parent and then use the identical chemistry, growth, fission, challenge, and observation schedule. Output-swap invariance gives $\mathbb{P}_\sigma^{erase}(A_\sigma(H, \epsilon_H)) \leq 1/2$. Hence the explicitly defined full-cycle causal contrast obeys

$$\boxed{\Delta_{H,\sigma}(N) := \mathbb{P}_\sigma^{int}(A_\sigma(H, \epsilon_H)) - \mathbb{P}_\sigma^{erase}(A_\sigma(H, \epsilon_H)) \geq \frac{1}{2} - HC_0 e^{-cN}.} \quad (61)$$

The same construction also gives an explicit comparison with the actual parent channel. Let $K_{\text{par},N} = \mathcal{K}(x_{N,\theta}^\sigma)$ be the pre-intervention channel of the balanced lattice parent, and put $d_* = d_K(K_\sigma^*, \rho K_\sigma^*) > 0$. Balanced rounding and the local Lipschitz bound for $\mathcal{K}$ give a constant $B_\theta < \infty$ such that

$$d_K(K_{\text{par},N}, K_\sigma^*) \leq B_\theta/N.$$

Consequently, the survival event in (59) implies

$$d_K(K_H, K_{\text{par},N}) \leq \epsilon_H + B_\theta/N.$$

For a fixed parent-relative tolerance $0 < e < d_*/4$, first use Lemma 5 to shrink the local neighborhoods so that $\mathcal{K}(D_{out}^\sigma)$ lies within distance $e/2$ of K_σ^* , and then take N large enough that $B_\theta/N < e/2$. The actual-parent ball of radius e and its output-swapped image are disjoint, since

$$d_K(K_{\text{par},N}, \rho K_{\text{par},N}) \geq d_* - 2B_\theta/N > 2e.$$

Define $A_{\text{par},N}(H, e) = \{K_H \text{ is defined and } d_K(K_H, K_{\text{par},N}) \leq e\}$, using the same pre-intervention $K_{\text{par},N}$ in both arms. Undefined readout channels are failures. Output-swap invariance gives an erased-arm probability at most $1/2$, while intact survival implies $A_{\text{par},N}(H, e)$. Thus

$$\mathbb{P}_\sigma^{int}(A_{\text{par},N}(H, e)) - \mathbb{P}_\sigma^{erase}(A_{\text{par},N}(H, e)) \geq \frac{1}{2} - HC_0 e^{-c(\theta)N}.$$

The positive constants and the lattice-parent lower threshold may be reselected for the smaller neighborhoods before N and H are chosen. This actual-parent event and the fixed-equilibrium-reference event in (61) need not coincide at finite N with the same tolerance.

This is a finite-horizon metastability theorem, not an absorbing heredity theorem.

Theorem 16 — a positive reset floor forbids permanent named-orientation fidelity at finite N

Let $\tau_{exit}^{N,\sigma}$ be the generation-skeleton exit time. Suppose that for every h , conditional on $\mathcal{F}_h$ and $\tau_{exit}^{N,\sigma} > h$, the next generation contains a declared symmetry-restoring reset event with probability at least $r_N > 0$, and conditional on that reset and the same history the probability of returning to the specified orientation at the next observation is at most $b < 1$. Define

$$p_N = r_N(1 - b).$$

Then

$$\boxed{\Pr(\tau_{exit}^{N,\sigma} > H) \leq (1 - p_N)^H, \quad \Pr(\tau_{exit}^{N,\sigma} = \infty) = 0, \quad E\tau_{exit}^{N,\sigma} \leq p_N^{-1}.} \quad (62)$$

For the complete growing process, a uniform floor follows directly from a **no-reaction growth cycle followed by an empty focal daughter**. Compactness of the post-fission set D_{out}^σ gives $B_0 < \infty$ such that the initial number of all partitioned internal objects is at most $B_0 N$. On a no-reaction path that number remains constant while $V_N(t) \in [N, 2N]$. Every internal propensity and each of the four Poisson import intensities is bounded by a constant times N on these paths. For fixed θ and T_g , their integrated total hazard is therefore at most $A_0 N$, uniformly over every history whose current state lies in D_{out}^σ .

The conditional probability of no reaction, **including no import**, is at least $e^{-A_0 N}$. Conditional on that event, unbiased fission puts every internal object in the sibling with probability at least $2^{-B_0 N}$. At the immediate post-fission observation the focal daughter is empty, so it lies outside the mature coding neighborhood and has no defined readout rows. It has already exited; no subsequent recovery assumption is needed. Thus the one-step exit probability has the history-uniform lower bound

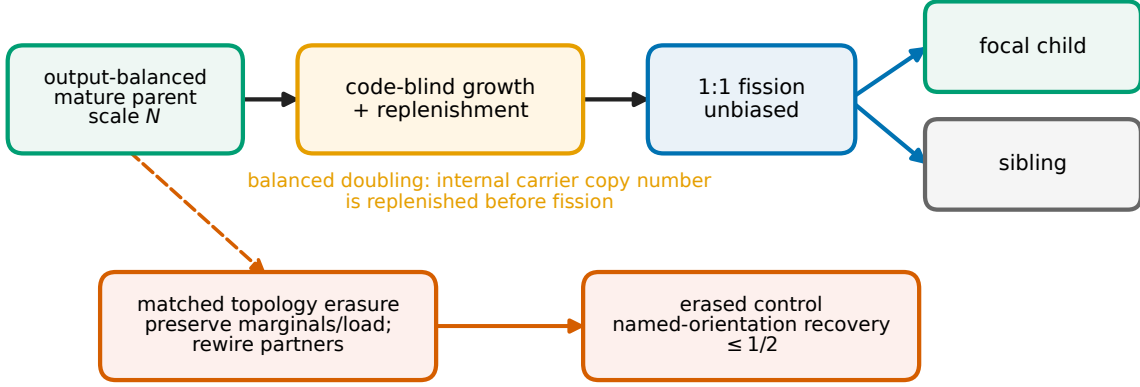
$$\begin{aligned} \Pr(\tau_{exit}^{N,\sigma} = h + 1 \mid \mathcal{F}_h, \tau_{exit}^{N,\sigma} > h) &\geq e^{-c_R N}, \\ c_R &= A_0 + B_0 \ln 2 > 0, \\ E\tau_{exit}^{N,\sigma} &\leq e^{c_R N}. \end{aligned} \quad (63)$$

The induction used for (62) applies with the direct exit floor $p_N = e^{-c_R N}$. It does not require a deterministic bound on all pre-fission counts along histories that survive only at observation times; Poisson imports would preclude that bound.

Thus heredity in a fixed finite compartment is metastable even when its characteristic lifetime grows exponentially with system size.

Figure 4 collects the neutral lineage mechanism, matched topology-erasure control, and the two finite-system bounds.

A Neutral lineage + endogenous replenishment



B Finite-system theorem boundaries

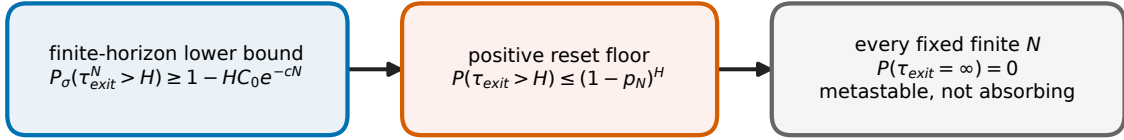


Figure 4: **Heredity and finite-system boundaries.** **A**, code-blind growth and internal templating/repair replenish carrier copy number before unbiased 1:1 fission; the matched topology-erasure control preserves declared marginals/load while randomizing cross-domain partner correlation. For the declared exact-output-balanced mature parents, the erased control recovers a named orientation with probability at most 1/2. **B**, finite-horizon lineage survival has an exponential-in- N lower bound on the declared compact regime, whereas any positive symmetry-restoring reset floor implies eventual loss of a fixed named orientation at every finite N .

7.5 Genesis-to-heredity compatibility and the end-to-end scope

The same reaction grammar supports both endogenous genesis and lineage heredity, but the two probability statements condition on different starting states. C1 proves that, from STRICT-NC, the growing elementary process has a strictly positive finite-horizon probability of entering a coding organization. C5 proves exponentially reliable finite-horizon heredity from a deliberately constructed, exact-output-balanced mature lattice parent. Their conjunction establishes **compatibility in one model family**, not a high-probability historical theorem from an arbitrary STRICT-NC realization to a long-lived lineage.

In particular, the present theorem does not supply a lower bound of the form

$$\Pr_{\text{STRICT-NC}}(\tau_{\text{mature}} \leq T_G, \tau_{\text{exit}} > H) \geq 1 - \varepsilon.$$

Nor does it claim that a generic stochastic parent produced by genesis is exactly output-balanced. Exact balance is used to obtain the sharp erased-control ceiling 1/2. For an unbalanced parent the appropriate null baseline is generally $1/2 + \delta_{\text{bal}}$, where the imbalance correction remains to be bounded.

These are not hidden assumptions of the current theorem; they are two stronger open problems:

1. **End-to-end probability problem:** quantify the probability and waiting time for STRICT-NC trajectories to enter the mature heredity core before extinction or reset.
2. **Unbalanced-parent comparator problem:** derive a continuity bound for the erasure baseline as a function of retained output-margin imbalance.

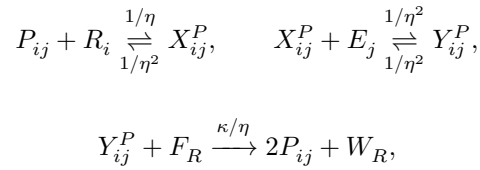
A stochastic numerical supplement should report both quantities, but numerical evidence would extend rather than repair the present analytic result.

8. Parasites, physical error, and the joint existence theorem

8.1 A local free-rider class

A parasite is defined by local chemistry, not by whether an analyst judges its mapping “wrong.” Parasite susceptibility and resistance are longstanding issues for hypercycles and other autocatalytic organizations [Eigen and Schuster 1979; Boerlijst and Hogeweg 1991]; the present free rider is a model-specific local stress test, not a claim to reproduce those systems. Consider a relation-like species that consumes the same receptor/effector building blocks and relation-copying fuel, can use the host-generated maintenance currency for repair, but contributes no decoded output to maintenance production. It obeys the same label-blind rule family for every i, j .

For the final elementary robust model this local free rider is realized explicitly by



Here κ/η is the effective first-order conversion coefficient after absorbing the fixed positive chemostatted F_R activity into κ ; that activity is not multiplied in a second time. The corresponding elementary bimolecular coefficient before this absorption is $\kappa/(\eta F_R)$. The fuel species remains displayed to record material and resource consumption.

This is supplemented by local damage, repair and turnover $P_{ij} \rightarrow Z_{ij}$, $Z_{ij} + M \rightarrow P_{ij} + W_M$, $P_{ij} \rightarrow R_i + E_j$, and $Z_{ij} \rightarrow R_i + E_j$, generated uniformly over all pairs. The label-blind physical-error test adds $A_{ij} \xrightarrow{\nu} P_{ij}$ with the same ν for every pair. Together with the four explicit growth-material imports, the final robust growing grammar has 115 instantiated reaction channels: 71 host channels, four imports, and 40 parasite/error channels. The fixed-volume host genesis assay is the separately defined host-only restriction with $g = 0$, containing 71 channels.

The error conversion $A_{ij} \rightarrow P_{ij}$ preserves both ports and removes the readout contribution. The robustness result covers this declared error family and admissible parameter perturbations; it does not establish tolerance to arbitrary wrong-output reactions, heterotypic binding, or topology-copying errors.

Near a parasite-free mature host state, each rare parasite pair mode has active/damaged linear block

$$L_P = \begin{pmatrix} \kappa s_g^2 - (d_P + \delta_P + g) & \rho_P y_g \\ \delta_P & -(\ell_P + \rho_P y_g + g) \end{pmatrix}. \quad (64)$$

Let

$$h_P = \ell_P + \rho_P y_g + g.$$

Its largest eigenvalue is

$$\Lambda_P = \frac{a_P - h_P + \sqrt{(a_P + h_P)^2 + 4\rho_P y_g \delta_P}}{2}, \quad (65)$$

where

$$a_P = \kappa s_g^2 - (d_P + \delta_P + g).$$

Theorem 17 — sharp local free-rider invasion threshold

The parasite is locally subcritical exactly when

$$\boxed{\kappa s_g^2 < d_P + g + \delta_P \frac{\ell_P + g}{\ell_P + \rho_P y_g + g}.} \quad (66)$$

Thus reflexive maintenance does not automatically exclude all free riders. Above the threshold the parasite-free coding equilibrium is linearly unstable in the enlarged local-chemistry model.

For the matched free-rider case

$$d_P = d, \quad \delta_P = \delta, \quad \ell_P = \ell, \quad \rho_P = \rho,$$

the host equilibrium identity collapses (66) to

$$\boxed{\Lambda_P < 0 \iff \kappa < bu_g, \quad \kappa_c(g) = bu_g.} \quad (67)$$

Combining this with (56) eliminates the latent branch amplitude:

$$\boxed{\kappa_c^2(1 - C^2) = 4\alpha b.} \quad (68)$$

8.2 Finite- η free-rider lifting, physical error, and division

The exact threshold (66)–(68) is a reduced/effective-model statement. To lift a **strictly subcritical** reduced margin to the explicit elementary parasite grammar, define the active parasite-unit total for one rare pair by

$$p = P + X^P + Y^P, \quad X^P = \eta \bar{X}, \quad Y^P = \eta \bar{Y}.$$

On fast time $\tau = t/\eta^2$, with host resources held fixed to leading order, the fast binding subsystem has leading Jacobian

$$A_f(E) = \begin{pmatrix} -(1+E) & 1 \\ E & -1 \end{pmatrix}, \quad \text{tr } A_f = -(2+E) < 0, \quad \det A_f = 1.$$

It is therefore uniformly attracting on every compact physical resource range. Its unique critical manifold is

$$\bar{X} = pR, \quad \bar{Y} = pRE.$$

Summing the P, X^P, Y^P equations cancels all binding/unbinding fluxes. Restriction to this critical manifold gives exactly

$$\dot{p} = [\kappa RE - (d_P + \delta_P + g)]p + \rho_P MZ, \quad \dot{Z} = \delta_P p - (\ell_P + \rho_P M + g)Z.$$

At the balanced reduced host branch $R = E = s_g$, $M = y_g$, this is precisely the block (64). Fix reduced macro parameters and $g \in (0, g_*)$, and let $E_H^{(\eta, g)}$ denote the stable host equilibrium family supplied by Theorem 14 for $0 < \eta < \eta_H(g)$. Assume the reduced parasite block $L_P^{\text{red}}(g)$ has strict dominant-eigenvalue margin

$$\Lambda_P^{\text{red}}(g) \leq -m_P < 0.$$

Normal-hyperbolic persistence and spectral continuity then give an $\eta_P(g) > 0$ such that every

$$0 < \eta < \min\{\eta_H(g), \eta_P(g)\}$$

explicit elementary parasite extension is locally subcritical. The exact identity $\kappa_c = bu_g$ is not asserted as an exact finite- η threshold; only persistence of the strict reduced margin is used.

At $\nu = 0$ the parasite-free full Jacobian is block triangular after host variables are ordered before parasite variables, because parasite-row derivatives with respect to host perturbations vanish when all parasite coordinates are zero. The host block is hyperbolically stable and the finite- η parasite block is strictly stable. For each fixed η in the range above, let D_η be the parasite-coordinate linear block in $q = (P, Z, X^P, Y^P)$. The elementary reaction grammar gives $D_\eta = \text{diag}_{ij} D_{\eta,ij}$, four 4×4 Hurwitz Metzler blocks on $(P_{ij}, Z_{ij}, X_{ij}^P, Y_{ij}^P)$. Each pair block is irreducible because positive host resources and reverse, damage, and repair rates connect its four coordinates; the full 16×16 matrix is reducible. The uniform error $A_{ij} \rightarrow P_{ij}$ supplies a source νs_{ij} with strictly positive P_{ij} component in **every** pair block, since every $A_{ij}^* > 0$. The implicit-function branch therefore satisfies

$$q'(0) = -D_\eta^{-1} s, \quad -D_\eta^{-1} = \int_0^\infty e^{D_\eta t} dt \geq 0.$$

Applying irreducible Metzler positivity separately to each source-bearing pair block gives $q'(0) > 0$ in all sixteen coordinates. Hence, for every such fixed η , there exists $\nu_0(g, \eta) > 0$ such that $0 < \nu < \nu_0(g, \eta)$ yields a **physical** nearby robust equilibrium with strictly positive parasite coordinates,

$$P, Z, X^P, Y^P = O(\nu),$$

while $\nu = 0$ gives the parasite-free boundary state. All previously positive host concentrations and strict coding/reflexive-maintenance margins persist by continuity.

The robust equilibrium remains in the simultaneous row/column-swap checkerboard subspace. At the post-fission reset volume N , the target receptor and effector moiety counts are exactly N under $C_R^* = C_E^* = 1$. Construct an **exact-balanced mature lattice parent** $x_{N,\theta}^\sigma$ by rounding every symmetry-paired class concentration in equal swapped pairs (checkerboard entries are rounded as $[[P_{c,N}, Q_{c,N}], [Q_{c,N}, P_{c,N}]]$) and rounding the remaining output-indexed nuisance coordinates in equal pairs. Define the free R_i, E_j counts as the exact integer residuals required by those target totals. Because the limiting equilibrium has strictly positive free material, these residuals are nonnegative for all sufficiently large N . The concentration error is $O(1/N)$, while the retained output-side margins are exactly swap-invariant. Because the robust equilibrium is an interior point of the declared coding basin relative to $\mathcal{X}$, including its parasite-free face when $\nu = 0$, $x_{N,\theta}^\sigma \in D_{in}^\sigma(\theta)$ for all sufficiently large N .

For causal-heredity comparisons in this robust state, extend the erasure intervention to **every** pair-bearing class in the final model,

$$\mathcal{C}_{\text{pair}}^{\text{full}} = \{B, A, D, U, H, X, Y, P, Z, X^P, Y^P\}.$$

Classwise receptor/effector multiplicities are constant across (i, j) , so the same uniform slot-bijection kernel preserves the appropriate weighted receptor/effector marginals, class totals and burden. The product kernel $\mathcal{E}_{\text{full}}$ satisfies

$$\mathcal{E}_{\text{full}}(\rho x, \rho A) = \mathcal{E}_{\text{full}}(x, A).$$

The small-error robust checkerboard equilibrium remains in the simultaneous row/column-swap fixed subspace, so every retained class margin is output-swap balanced. Thus the erased robust-parent law is exactly output-swap invariant and the specified-orientation descendant probability remains at most $1/2$ under the complete robust chemistry, explicit growth boundary and single-lineage fission.

Freeze the host at its deterministic mature equilibrium and consider the reduced rare-parasite first-moment linearization. During one volume-doubling cycle the expected absolute rare-parasite vector acquires a volume factor $e^{gT_g} = 2$, followed by the mean focal-lineage partition factor $1/2$. These cancel, so neutral

growth and fission preserve the local invasion sign. The principal multiplier of the resulting mean linear cycle map is

$$\boxed{\mathcal{M}_P^{cycle} = \text{spr}(e^{L_P T_g}) = e^{\Lambda_P^{\text{red}}(g) T_g}.} \quad (69)$$

Here spr denotes spectral radius. This multiplier describes the dominant linear mode at the frozen mature host; an arbitrary initial parasite composition need not have the same total-count ratio over a cycle. It is not an almost-sure multiplier for a realized low-copy lineage in a fluctuating host. For the finite- η elementary block, the corresponding first-moment invasion exponent is $\Lambda_{P,\eta} = \Lambda_P^{\text{red}}(g) + O(\eta)$, so the same **negative sign** persists for sufficiently small η when the reduced margin is strict. No exact finite- η threshold identity is claimed.

8.3 Main theorem

Main Theorem — conditional existence of operational molecular coding without a pre-existing cross-domain decoder

Fix the compartment protocol: volume doubles from N to $2N$ over $T_g = \log 2/g$, then independent unbiased Bernoulli $1/2$ molecule partition selects a tracked daughter of reset volume N . Fix also the material normalization $C_R^* = C_E^* = 1$. Let $\mathcal{P}_{adm}$ denote the space of **microscopic elementary reaction rates, positive boundary activities, positive growth rate g , η , and error/free-rider parameters**, subject to these fixed model choices, the declared type/output-symmetry equalities and nonnegative-rate constraints, and endowed with its relative topology. The partition probability, doubling protocol and material normalization are fixed model data, not freely perturbable coordinates. There exist a fixed label-blind local reaction/compartment grammar, a binary post-hoc assay interface, and an admissible microscopic parameter point θ_* obtained in the order

$$\begin{aligned} \text{reduced macro parameters} &\xrightarrow{\text{realization (39b)}} \lambda_* \\ &\rightarrow g \in (0, g_*) \\ &\rightarrow 0 < \eta < \min\{\eta_H(g), \eta_P(g)\} \\ &\rightarrow 0 \leq \nu < \nu_0(g, \eta). \end{aligned}$$

with a strictly subcritical reduced local free-rider margin. Because all required stability, coding, reflexive, and invasion margins at θ_* are strict, there exists a nonempty **relatively open** symmetry-respecting neighborhood $U \subset \mathcal{P}_{adm}$ of θ_* and a nonempty compact core $U_0 \Subset U$ on which those strict properties persist. For every full parameter vector $\theta \in U_0$:

1. **STRICT-NC origin.** The initial law is the explicit μ_0^N in (27). No pair-bearing carrier is initially present. The typed grammar contains no pre-existing decoder-equivalent cross-domain relation, no direct $S_i \rightarrow O_j$ rule, no active cross-domain interpreter, no target mapping, and no code-quality reward, and it satisfies the declared material/resource/energy boundary. Domain-local symbol/receptor recognition and effector/output actuation are primitive, independently relabelable port operations; the only initial cross-domain assembly is the uniform Cartesian-product schema.
2. **Endogenous readout genesis.** Local dynamics constructs cross-domain relation states from factorized receptor and effector components. For every sufficiently large finite N , Proposition 1 gives the final elementary growing process strictly positive probability to enter a coding organization from μ_0^N before its first fission. Output-swap equivariance makes the two strict first-hit probabilities equal; conditional on a hit of their union by the declared horizon, each orientation has probability $1/2$. Separately, the host-only fixed-volume elementary genesis CTMC in Theorem 10A satisfies the stronger result of almost-sure union hitting and unconditional exact $1/2$ – $1/2$ first-hit.
3. **Operational coding.** The constructed relation mediates a reproducible intervention-defined symbol-conditioned output channel with $C(K) > 0$. Matched relation/interpreter ablation either destroys the required channel by driving required total output flux to zero, in which case no post-ablation normalized K is assigned, or—if a normalized ablated channel remains defined—strictly lowers the declared specificity/fidelity. The same symbols can realize the opposite mapping in the symmetry-related organization.
4. **Local causal consequence.** Matched disruption of the realized readout path reduces a physically represented code-mediated maintenance flux. In the finite- η elementary assay, both arms first replace

inherited active outputs by equal-moiety inert isomers, after which only the disabled arm redirects new readout products; the strict contrast is (26a). No mapping-level fitness score is consulted.

5. **Persistence.** The coding organization survives nonzero turnover/noise for nontrivial finite horizons, with residence increasing with system scale in the stated finite-system regimes.
6. **Causal heredity.** For each orientation $\sigma \in \{+, -\}$ and every sufficiently large N , choose the **exact-output-balanced post-fission mature lattice parent** $x_{N,\theta}^\sigma \in D_{in}^\sigma(\theta)$ obtained by symmetry-preserving checkerboard rounding of the corresponding robust equilibrium, with every retained output-indexed class margin fixed by the output swap. Under the generation skeleton and the intact/erased measures defined in Section 7.1, there exist finite positive constants $C(\theta), c(\theta)$ and $N_0(\theta)$ such that, for every $N \geq N_0(\theta)$ and every integer lineage horizon $H \geq 2$ satisfying

$$HC(\theta)e^{-c(\theta)N} < \frac{1}{2},$$

$\Delta_{H,\sigma}(N) > 0$: descendants retain a strictly positive parent-relative same-channel advantage over the matched **full-state** topology-erasure baseline, whose specified-orientation probability is at most $1/2$ for these exact-balanced parents. This includes at least two complete parent-to-descendant transmissions. In particular, for every $0 < \gamma < c(\theta)$, any sequence $H_N \leq e^{\gamma N}$ has lineage survival probability tending to one. No $1/2$ erasure ceiling is asserted here for arbitrary unbalanced basin parents.

7. **Reflexive interpreter maintenance.** Code-mediated outputs causally support future capacity of the same relation/interpreter system through explicit local maintenance/repair chemistry, with all relevant reaction rules present from the STRICT-NC origin.
8. **Open robustness.** The foregoing deterministic and finite-horizon stochastic properties coexist on the compact core U_0 contained in the relative-open neighborhood U for the single explicitly assembled elementary model with the symmetry-neutral growth-material boundary. The exact identities (55), (67), and (68) belong to the reduced/ $\eta = 0$ model. If $L_P^{\text{red}}(g)$ has a strict negative invasion margin, then every $0 < \eta < \min\{\eta_H(g), \eta_P(g)\}$ in the stated neighborhood has a locally subcritical elementary parasite extension. For each such fixed η , sufficiently small $0 \leq \nu < \nu_0(g, \eta)$ gives a physical nearby coding equilibrium; for $\nu > 0$ the parasite coordinates are strictly positive and $O(\nu)$. The full pair-bearing erasure kernel $\mathcal{E}_{full}$ preserves the causal-heredity comparator for the exact-balanced robust parents specified in item 6.

Therefore one fixed model family supports **positive-probability genesis of a contingent cross-domain operational assignment/coding relation from the specified STRICT-NC state**, and, separately from the stated exact-balanced mature-parent class, **causal heredity and reflexive interpreter maintenance**. This is a compatibility theorem, not an unconditional high-probability origin-to-lineage history.

Proof dependency and quantifier order. Items 1–3 follow from Sections 3–4, the explicit macro-to-micro realization (39b), Theorems 10 and 10A, and Proposition 1. Item 4 is Corollary 2 including its finite- η intervention. Item 7 is Theorems 7–8 and is preserved by Theorem 10, whose normal-hyperbolicity hypothesis is discharged in Appendix E. Theorem 11 supplies only the separate two-fission pure-partition assay. Items 5–6 for complete growth/reaction/fission cycles follow from Theorems 14–16, the stopped-process concentration result in Appendix F, and the symmetry-neutral material boundary of Section 7.3. Item 8 follows from Theorems 10 and 17 together with Section 8.2 and Appendix G. The construction order is explicit: choose the reduced macro parameters, construct λ_* by (39b), choose $g \in (0, g_*)$ and a strict subcritical free-rider parameter, then choose $0 < \eta < \min\{\eta_H(g), \eta_P(g)\}$, and finally $0 \leq \nu < \nu_0(g, \eta)$. Fix any positive assay duration τ before choosing N, H ; the controlled-readout normalization is exactly independent of that duration. These choices determine the microscopic center θ_* , after which U and U_0 are chosen by continuity of strict margins in the actual elementary rate/activity vector. For each $\theta \in U_0$, choose $D_{in}^\sigma(\theta) \subseteq D_{out}^\sigma(\theta) \subseteq K^\sigma(\theta)$, then N , then H . No uniform common basin over all U_0 is asserted. The continuous parameter vector is fixed before the large- N stochastic limit; no joint limit $\eta \rightarrow 0$ with $N \rightarrow \infty$, no fixed-time large- N branch-selection claim from exactly symmetric data, and no permanent named-orientation heredity claim at fixed finite N is made. The neighborhood U is open in the **relative topology of the microscopic space** $\mathcal{P}_{adm}$; exact symmetry equalities, nonnegative-rate constraints, the fixed unbiased doubling/partition protocol, and the unit material normalization are part of the model class rather than additional perturbation directions. $\square$

Claim type. This is a conditional mathematical model-existence theorem. It is not a theorem about the historical origin of terrestrial life, the origin of the Standard Genetic Code, or the prebiotic availability of the model’s molecular implementation.

8.4 Representative computational certificate

For one non-boundary reduced-model point,

$$\alpha = 1, \quad b = 16, \quad d = 1.5, \quad \delta = 0.1, \quad \beta = 1, \quad \rho = 2, \quad \ell = 0.5, \quad \mu = 0.7,$$

the representative reflexive equilibrium has approximately

$$u = 0.5794371945, \quad p = 0.4361319354, \quad q = 0.1433052591,$$

$$C = 0.5053639620, \quad \max \Re \lambda(J) = -0.2598875203.$$

The matched-output-block leading capacity coefficient is approximately $0.0096217374 > 0$. Under neutral growth $g = 0.1$, the matched free-rider threshold is approximately

$$\kappa_c = 8.8900283954.$$

These computations are implementation certificates only. They do not replace the analytic arguments above and do not define universal biochemical constants.

A separate **single joint finite- η witness** checks that the principal robustness clauses are not supported only by mutually incompatible representative points. Using the same continuous parameter tuple

$$(\alpha, b, d, \delta, \beta, \rho, \ell, \mu, g, \eta, \kappa, \nu) = (1, 16, 1.5, 0.1, 1, 2, 0.5, 0.7, 0.1, 5 \times 10^{-4}, 8, 10^{-5}),$$

the elementary realization takes $K_B = K_H = K_R = K_E = K_S = 1$, all five associated reverse-rate scales equal to 1, and dimensionless boundary activities $F_A = F_R = Q_* = S_* = F_M = 1$. Equation (39b) then gives $k_A = 1$, $k_{cat} = 16$, and $k_Q = 1$; the witness uses $k_M = 1$. The parasite receptor/effector capture and release coefficients are 1 in the scaled convention, $(d_P, \delta_P, \rho_P, \ell_P) = (1.5, 0.1, 2, 0.5)$, and $C_R^* = C_E^* = 1$. Thus its induced slow coefficients are exactly $(\alpha_{eff}, b_{eff}, \beta_{eff}) = (1, 16, 1)$ at $\eta = 0$, rather than being inferred from an unrelated microscopic tuple.

An explicit 51-dimensional elementary-host + free receptor/effector material species + symmetry-neutral growth imports + matched-free-rider + positive label-blind-error system has equilibrium residual 2.39×10^{-12} , operational specificity $C = 0.40655$, minimum free receptor/effector concentrations above 0.431, total receptor and effector moiety concentrations equal to the target value 1 to numerical precision, full-Jacobian spectral abscissa approximately -0.1000 (the new material-return modes occur at the exact scale $-g$), and a strictly positive matched support-block contrast at every tested time from 0.05 to 1. The total parasite-like load is approximately 8.15×10^{-5} . The reduced matched-free-rider threshold at the same macro point is $\kappa_c = 8.89003 > 8$. This is a **computational joint-witness certificate only**; stochastic N, H bounds are established in Appendix F rather than inferred from a fitted finite-population threshold.

9. Falsifiable causal discriminators

A model-existence theorem is more useful if it specifies interventions that distinguish its mechanism from explicit alternatives. We therefore separate generic robustness signatures from mechanism-class discriminators.

9.1 Signals that are not specific enough on their own

A growth/dilution threshold at which an organized state disappears is not, by itself, evidence of molecular coding: ordinary autocatalytic and multistable chemical networks can show comparable boundaries. Likewise, exponential transition-time scaling between attractors is a generic large-deviation/small-noise phenomenon for broad classes of chemical Markov jump processes under the requisite hypotheses [Agazzi, Dembo and Eckmann 2018]. These observables remain consistency checks for the present model, but they are not standalone coding-specific discriminators.

9.2 Coupled specificity–parasite boundary

For the **reduced/effective matched free-rider realization**, (68) predicts

$$\kappa_c^2(1 - C^2) = 4\alpha b. \quad (70)$$

Across conditions that preserve α and b , this gives the parameter-free ratio

$$\frac{\kappa_{c,1}}{\kappa_{c,2}} = \sqrt{\frac{1 - C_2^2}{1 - C_1^2}}. \quad (71)$$

Violation of (70)–(71), after verifying the matched kinetic assumptions, falsifies this particular matched-free-rider realization. It does not falsify every possible model satisfying the operational criteria defined here.

9.3 Topology-erasure null theorem

Let $S_{comp}(X)$ collect the domain-local marginals, structural-class totals, bulk resources, and declared energetic burden that a matched topology-erasure intervention preserves. Call a comparator **composition-sufficient** when the future channel/heredity law depends on X only through $S_{comp}(X)$, not on the erased receptor–effector partner correlation.

Theorem 18 — composition-sufficient systems have zero matched topology causal contrast

If

$$S_{comp}(X^E) = S_{comp}(X),$$

then for every declared future assay horizon,

$$\mathcal{L}(Y_H | X) = \mathcal{L}(Y_H | X^E), \quad \Delta_{top}^{comp} = 0. \quad (72)$$

By contrast, the intact model lineage satisfies (61) on the declared compact coding regime. Thus a positive matched topology-erasure effect separates topology-dependent causal heredity from the composition-sufficient null class.

9.4 Why a two-axis causal fingerprint is insufficient

Positive topology-heredity and positive structural self-support do not by themselves prove an operational code. A relational autocatalytic network with no symbol-conditioned readout can depend on its partner topology and can produce generic support for that topology. Therefore an ungated pair

$$(\Delta_{top}, \Delta_{support}) > (0, 0)$$

is not sufficient for coding.

The admissible sequence is instead

$$G_{code} \longrightarrow \Delta_{top} \longrightarrow \Delta_{ref}. \quad (73)$$

The **code gate** G_{code} independently requires a challengeable symbol-conditioned channel with positive specificity and a matched interpreter ablation that either drives a required total output flux to zero, leaving the normalized ablated channel undefined, or strictly decreases the declared specificity/fidelity when that channel remains defined, as in Section 2.3. Only after this gate is passed is the support intervention interpreted as reflexive interpreter maintenance.

9.5 Imperfect matching and quantitative separation

Let the ideal causal margins for topology and reflexive support be $m_j > 0$, $j \in \{T, R\}$. Let ϵ_j quantify the physical mismatch of the corresponding matched intervention, let $L_j\epsilon_j$ bound the largest null response attributable to nuisance mismatch, and let $K_j\epsilon_j$ bound the reduction of the ideal model margin under implementation mismatch.

Theorem 19 — robust matched-intervention separation

If

$$m_j > (K_j + L_j)\epsilon_j, \quad (74)$$

then the threshold

$$\tau_j = \frac{L_j\epsilon_j + m_j - K_j\epsilon_j}{2} \quad (75)$$

separates the declared null and model prediction classes, with common margin

$$\Gamma_j = \frac{m_j - (K_j + L_j)\epsilon_j}{2} > 0. \quad (76)$$

A system that passes G_{code} and has positive corrected topology and reflexive margins therefore rejects the explicit union of (i) composition-sufficient heredity and (ii) already-coded but non-reflexive comparators, within the declared calibration bounds. This identifies an operational mechanism class rather than a unique microscopic chemistry.

Figure 5 combines the exact specificity–free-rider relation with the gated falsification protocol.

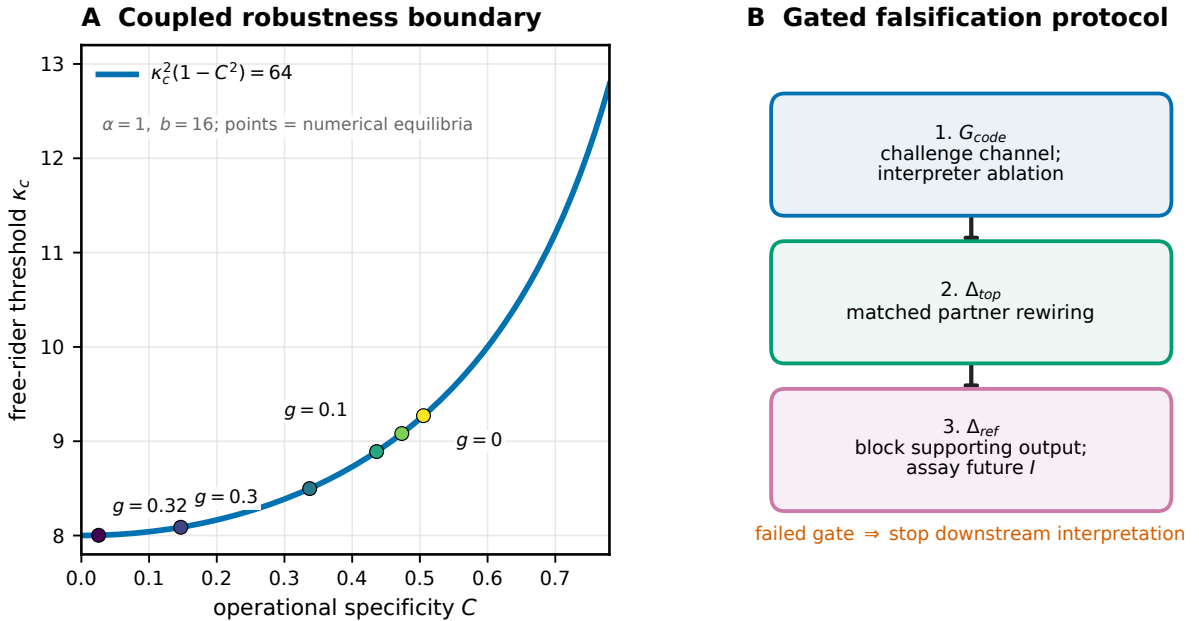


Figure 5: **Robustness and falsification.** **A**, the reduced matched free-rider family obeys the exact relation $\kappa_c^2(1 - C^2) = 4ab$; the curve and points shown use the representative family $\alpha = 1, b = 16$, with points reproduced by the accompanying reduced-equilibrium calculation. **B**, experimental interpretation is gated: an operational code gate must pass before topology-heredity and reflexive-maintenance contrasts are assigned those meanings.

9.6 Experimental implementation logic

Programmable strand-displacement/exchange chemistry offers a high-control proof-of-principle platform for matched partner rewiring, channel challenge, and fluorescence-based readout [Zhang and Winfree 2009; Kim et al. 2020]. Such an engineered circuit can test the **causal intervention logic** of (72)–(76). If the symbol-to-output relation is explicitly programmed into the DNA design, however, it cannot test decoder-free genesis; that would violate the STRICT-NC origin requirement.

Autocatalytic RNA networks in droplets or coacervates provide a closer comparator for prebiotic self-organization and compositional heredity [Ameta et al. 2023; Pavlinova et al. 2023]. A full assay of the type defined here on such a platform would still require: (i) an independently challengeable symbol-conditioned channel, (ii) a relation-topology rewiring intervention that preserves matched composition/load, and (iii) a decoded-output support intervention that tests future interpreter capacity. Existing autocatalysis or compositional persistence alone is not sufficient.

10. Discussion

10.1 Established properties and interpretive scope

The most stable statement of the result is vocabulary-neutral: the model contains a **contingent, remappable, causally mediated cross-domain channel** that is physically assembled, supports two symmetry-related assignments, affects descendant channels through inherited relation topology, and receives causal maintenance support from its own outputs. The term *operational molecular code* refers to the channel under the intervention and specificity criteria of Section 2. Heredity and output-mediated maintenance are additional causal properties, assessed separately.

Three causal edges remain logically independent:

$$\begin{aligned} \text{symbol challenge} &\rightarrow \text{output through interpreter,} \\ \text{relation topology} &\rightarrow \text{descendant channel,} \\ \text{code-mediated output} &\rightarrow \text{future interpreter capacity.} \end{aligned}$$

Their conjunction is the paper’s **Causal Closure Triangle** (Figure 3). None of the edges is inferred from a global semantic score, and the dynamics do not evaluate the triangle as an objective.

Within the stated model assumptions, the results support the following distinctions:

Property	Content	Status here
Contingent, remappable channel	The same neutral reaction grammar admits opposite assignments mediated by relation topology	Established
Operational coding	The channel meets the specified intervention and specificity criteria	Established under those criteria
Causal heredity	Intact descendants retain a channel advantage over matched topology erasure for the specified mature parents and finite horizons	Established under those conditions
Output-mediated maintenance	Readout products causally support future interpreter capacity through a common maintenance process	Established
Mapping-dependent maintenance	Maintenance depends on which symbol–output assignment is realized	Not established
Encoded interpreter construction	Encoded products specify or build the machinery that interprets the same descriptions	Not established

The table distinguishes the properties established by the model from additional properties not established here.

10.2 What the constructed relation explains

An ordinary switch can produce alternative states through a fixed state-dependent mechanism. The present construction adds four properties that are not generic to switches: independent symbol/output domain symmetry, physically contingent cross-domain assignment, remappability between opposite mappings, and causal transmission of the assignment topology across lineage events. These features support the operational-code terminology and place the result near contingent-mapping and semantic-capacity formalisms [Görlich and Dittrich 2013; Dittrich 2018].

The model nevertheless begins with domain-local recognition and actuation primitives. It therefore does not derive all discriminatory chemistry from an undifferentiated molecular soup. A fair physical description is an **initially unwired cross-domain architecture** whose relation topology is generated and selected endogenously. The theorem excludes a pre-existing assignment table; it does not exclude every local interface from which such a table can later be assembled.

This boundary is relevant to Code Biology and lower-threshold discussions [Lacková and Faltýnek 2021; Barbieri 2025]. The present model provides contingency, remappability, mediation, repetition, and a restricted heredity criterion. It does not claim to decide whether those conditions are sufficient for representation, interpretation, or semiosis in a broader sense.

10.3 Output-mediated maintenance and further closure questions

Theorem 8 proves that code-mediated outputs causally increase future interpreter capacity through a common maintenance currency. This is a counterfactual loop: disabling the readout path while matching immediate burden removes a positive maintenance flux and lowers future capacity.

The loop is nevertheless **mapping-insensitive at the maintenance stage**. Both output classes can feed M , so the system does not require a specific output identity to rebuild a matching interpreter. The result does not establish the interpreter-construction property illustrated by artificial-chemistry closure models and by biological translation, in which encoded products help build the machinery that interprets the same encoded descriptions [Sharov 2009b; Wills 2009; Clark, Hickinbotham and Stepney 2017].

This distinction identifies two further research questions: under what conditions does maintenance depend on the realized assignment, and under what conditions do encoded products specify or build the interpreter? The present result does not prescribe a necessary order for resolving them.

10.4 Genesis, mature-parent conditioning, and end-to-end scope

The theorem establishes positive-probability genesis from STRICT-NC and high-probability finite-horizon heredity from exact-balanced mature parents in the same model family. It does not prove that a generic STRICT-NC realization reaches that selected parent class with high probability, nor does it provide an unconditional waiting-time law for the complete origin-to-lineage sequence.

This distinction matters for interpretation. The mathematical conclusion is that genesis, maturation, reflexive support, and heredity are **jointly compatible** under one fixed reaction grammar and one nonempty parameter regime. It is not yet an end-to-end origin history. A stronger theorem would combine first-passage to the mature core with the subsequent lineage survival event and would quantify the dependence on N , g , η , and resource activity.

The exact-balanced parent is similarly a deliberate identification device. It produces an exact swap-invariant erasure null and therefore a sharp $1/2$ ceiling. The theorem does not extend that exact ceiling to arbitrary stochastic parents. A quantitative imbalance correction would widen the empirical scope and is a natural next mathematical target, but it is not required for the present existence result.

10.5 Boundary theorems are part of the result

Several negative results materially constrain the positive construction.

First, causal chemical activity alone does not constitute coding: if symbol identity is erased before relative output selection, the operational channel is exactly non-specific. Second, symbol-side memory without

an output-relational state remains insufficient under output symmetry. Third, in the shared-resource model, linear or concave local reinforcement cannot destabilize the assignment-symmetric state; sufficient superlinearity or a different mechanism is required. Fourth, finite closed fuel cannot sustain perpetual reflexive support. Fifth, a fixed finite ancestral memory stock cannot support indefinite above-baseline causal heredity under passive unbiased segregation. Sixth, every positive finite-system reset floor destroys permanent fidelity of a named orientation. Seventh, reflexive maintenance does not guarantee resistance to arbitrarily strong free riders. Finally, positive topology and support contrasts are not evidence of coding unless an operational code gate is independently passed.

These failures prevent the positive result from being read as a generic claim that “autocatalysis becomes semantics.” Table 2 collects the corresponding boundary statements.

10.6 Relation to prior coding and closure theories

Self-organization of genetic coding, assignment catalysts, physical interpreters, reflexive information, semantic closure, molecular-code motifs in reaction networks, and compositional heredity all have a substantial body of prior work. The present work therefore does not claim originality for any of these ingredients separately.

The comparison is sharper when the operative assumptions are separated. Noisy-channel optimization models commonly assign a code-level objective involving accuracy, diversity, or cost [Tlustý 2007, 2008; Piraveenan, Polani and Prokopenko 2007]. The present construction excludes mapping-quality terms from the dynamics. Molecular-code and semantic-capacity formalisms ask whether a medium can implement contingent mappings [Görlich and Dittrich 2013; Dittrich 2018]; the present construction adds endogenous relation genesis, matched mediation, and lineage intervention. Code Biology and lower-threshold work emphasize adaptor relations, arbitrariness, representation, repetition, historicity, and self-replication [Lacková and Faltýnek 2021; Barbieri 2025]; the present analysis establishes only the subset encoded by its operational criteria. Semantic-closure accounts emphasize a code plus an interpretation system [Sharov 2009b; Wills 2009]; the present construction establishes the output-mediated maintenance property described above.

A focused comparison clarifies the added question. The reaction-network capacity analysis [Görlich and Dittrich 2013] identifies contingent mappings available in a medium; here a dynamical construction is coupled to readout and lineage interventions. A recent protocell model [Nunes Palmeira et al. 2026] studies the evolution of growth and informational functions with rudimentary copying and translation specified; here the lineage protocol is neutral with respect to assignment quality, and the measured target is the causal effect of pairing topology. These differences concern explanatory targets and assumptions, not a ranking of realism or a claim that the present model solves what those studies leave open.

The distinctive contribution defended here is the explicit compatibility of relation formation, remappable readout, conditional causal heredity, and output-mediated maintenance within a specified recognition architecture, together with matched discriminators and finite-system boundaries. No individual ingredient is claimed as unprecedented. Whether an earlier construction establishes the same conjunction at comparable assumptions remains a question for literature comparison; a matching precedent would narrow the originality claim without invalidating the model’s mathematics.

10.7 Finite systems, stochastic choice, and metastability

The word *spontaneous* is used only in a finite stochastic symmetry sense. From the exactly symmetric initial state, finite trajectories first hit the two symmetry-related coding basins with equal probability. If the large-system deterministic limit is taken first over a fixed time interval, exact symmetry is preserved and no branch is selected. At arbitrarily long times every finite irreducible realization may also mix between orientations. The operational object is therefore a stopped, lineage-local coded organization with residence that grows with system scale, not a nonzero stationary signed mapping.

Likewise, causal heredity is metastable rather than permanently absorbing at fixed finite N . The model supports lineage horizons increasing exponentially with N under the stated compact-basin assumptions, while a positive reset floor still implies eventual loss of a fixed named orientation with probability one.

10.8 Chemical admissibility, engineering complexity, and historical plausibility

The elementary network supplies low-molecularity mass-action reactions, exact formal moiety conservation, explicit fuels/wastes, an explicit 115-channel inventory, and a controlled singular-limit connection to the reduced theorem. This is a chemical-admissibility statement at the chosen abstraction level. It is not a proof that the required species, rate hierarchy, chemostats, or modular reaction architecture were available on the early Earth.

The construction is tailored to its logical target. It is not demonstrated to minimize species count, reaction count, number of fuels, fine tuning, or synthetic implementation cost. The binary alphabets, exact type symmetry, fast-slow separation, sustained material boundary, and engineered repair modules make the current witness better suited to an existence theorem and a synthetic proof of principle than to a direct prebiotic scenario.

Nothing in the theorem identifies codons, amino acids, tRNAs, aminoacyl-tRNA synthetases, ribosomes, triplet coding, or the Standard Genetic Code. Mapping either branch to a modern biological assignment would add information that is absent from the model.

10.9 Empirical and computational tests

No simulator output or numerical trajectory is a premise of the theorem or conclusion. Auxiliary computations could nevertheless clarify the mechanism, expose quantitatively restrictive regimes, and guide experimental design. Five useful further analyses would be:

1. **Genesis:** first-passage-time distributions from STRICT-NC and an empirical test of the conditional $1/2-1/2$ orientation symmetry.
2. **End-to-end trajectories:** paths that enter the mature core, undergo growth/fission, retain the channel for multiple generations, and eventually reset.
3. **Scale dependence:** residence-time and lineage-survival curves versus N , with comparison to the proved exponential bounds.
4. **Comparator robustness:** intact-versus-erased descendant channels as a function of parent output-margin imbalance, estimating the correction to the exact $1/2$ null.
5. **Robustness surfaces:** stochastic free-rider invasion and error sensitivity near the reduced $\kappa_c(C)$ relation across (g, η, ν) .

These computations would not replace the analytic proofs. Their role would be to demonstrate representative effect sizes, expose finite-size regimes, and guide experimental design. A controlled strand-displacement implementation could test the intervention logic, while RNA/coacervate systems provide a nearby comparator for autocatalysis and compositional heredity [Ameta et al. 2023; Pavlinova et al. 2023]. Neither platform would by itself establish a historical origin.

10.10 Limitations and open directions

The current theorem is binary and model-specific. It does not establish a universal necessity of the CHRT mechanism, a universal critical surface for coding, or the minimal chemistry capable of meeting C0–C7. The exact quasipotential and switching prefactors of the complete stochastic elementary network remain uncomputed. The numerical growth/dilution loss point found for one parameter set is not a general theorem. The multi-cut copy-number product bound requires both independent exact cut survival and the joint failure baseline in Theorem 12. The experimental mismatch theorem requires externally calibrated nuisance-sensitivity bounds. The energetic treatment declares maintained chemical affinities rather than deriving a detailed geochemical network.

The exact operational channel is measured by the controlled side-sample protocol of Section 2.3, including bound-symbol release and suspension of non-readout reactions. It quantifies the readout available from the inherited accessible carrier pool. A transient channel with full chemistry continuing during a symbol challenge is a different observable; its equality to the carrier-ratio formula is not established. Realizing the ideal preparation and selective reaction controls in a laboratory also remains an experimental task.

The unresolved questions fall into two groups. Within the current architecture, natural extensions are higher alphabets, quantitative unbalanced-parent erasure bounds, and an end-to-end origin-to-heredity probability theorem. For the originating decoder problem, the central extensions are a lower-level derivation of recognition and composable readout, an account of homotypic templating without stipulating

its target-level function, mapping-sensitive maintenance, and encoded interpreter construction. Experimental implementations and a finite-affinity thermodynamic realization would address further physical requirements. Each extension requires its own proof or experiment and a separate assessment against prior work; none is included in the present result.

10.11 From assignment formation to the origin of interpreting capabilities

The construction permits a direct accounting of what is supplied and what is produced. The distinction is functional: an absent molecular complex can be assembled by rules whose recognition and conversion capabilities have already been specified.

Feature	Supplied by the model	Outcome established here
Recognition and readout	Receptor discrimination, effector actuation, and their composability	An assembled carrier mediates a contingent channel
Relation formation	Uniform assembly, positive nucleation, and homotypic reinforcement	Opposite symmetry-related assignments and positive-probability genesis
Lineage transmission	Neutral growth, replenishment chemistry, and unbiased partition	Conditional topology-dependent descendant-channel advantage
Continued operation	Output-to-maintenance conversion and repair reactions	Outputs causally increase future interpreter capacity

The resulting division of explanatory work is not a new general account of interpreter origins. The material implementation of a mapping and the emergence of systems capable of interpretation already motivate earlier analyses [Görlisch and Dittrich 2013; Wills 2009]. The present model contributes an explicit case in which the generated relation and the assumed capacities can be inspected separately.

An explanation has premises, but that fact alone does not show that it assumes its own conclusion. General interactions may permit a functional structure without supplying an organized instance of that structure. Conversely, declaring symbol recognition, selective output, and topology copying as primitive operations leaves those specific capabilities unexplained. Treating every enabling physical condition as an already installed decoder would make the claim true by definition and remove its ability to distinguish competing mechanisms. The present theorem establishes neither the natural origin of all interpreting capabilities nor a universal impossibility of explaining them.

A stronger origin account should make three matters assessable. First, it should specify which decoding function is absent from its starting state and which lower-level interactions are assumed, with a physical justification independent of the desired assignment. Second, it should show how a structure implementing that function is reached dynamically, distinguishing positive-probability reachability from typicality or an empirically plausible timescale. Third, interventions should test that the newly formed structure mediates the function and that an externally assigned target or observer’s readout convention has not supplied it. These are proposed explanatory requirements for further work, not a proved universal criterion or a claim of historical sufficiency. In particular, the physical justification and origin of the specialized capabilities remain unresolved in the present construction.

10.12 Conclusion

This work establishes a conditional conjunction: within one specified reaction architecture, a cross-domain assignment can form without an initial pairing table or mapping-quality reward, mediate a remappable channel, support its future operation through decoded outputs, and exhibit topology-dependent causal heredity under the declared mature-parent conditions. The common parameter regime, matched interventions, and quantitative failure boundaries make the result an explicit reference construction for testing coupled coding functions.

Its academic contribution is the joint dynamical construction and causal discrimination of those functions. The recognition, assembly, templating, and readout capabilities enabling the construction remain assumptions. Their origin is a further mechanistic problem, and acknowledging it neither proves that a decoder must be externally supplied nor establishes that every naturalistic explanation is circular.

The originating question can now be stated more precisely: **what model and evidence would show that an interpreting capability is produced by physical dynamics, rather than supplied as a specialized recognition, templating, or conversion operation?** Answering that question requires explaining the capabilities at a lower physical level as well as the assignment they make possible.

Boundary and literature tables

Table 2 — Overstated formulations and supported replacements.

Overstated formulation	Supported replacement
“The model discovers the correct mapping.”	Two symmetry-related mappings have identical physical status; no target map exists in the dynamics.
“The deterministic symmetric limit spontaneously selects a code.”	Conditional on reaching either basin, first-hit orientations are equally probable; the fixed-time fluid limit remains symmetric.
“Reflexive maintenance guarantees parasite resistance.”	A locally defined free rider has a sharp conditional invasion threshold.
“Finite compartments inherit a named code forever.”	Named-orientation heredity is metastable; a positive reset floor implies eventual loss at every fixed finite N .
“Topology and self-support prove coding.”	An independent operational code gate must precede topology and reflexive-support interventions.
“The chemical model is a prebiotic pathway.”	The model is mass-action/moiety/fuel admissible at the declared abstraction level; historical availability is not established.
“An initially unwired model explains the origin of every decoding capability.”	It generates a relation within supplied recognition, templating, and readout capabilities; their origin is unresolved.
“Every origin explanation necessarily assumes a decoder.”	The present model provides no universal impossibility theorem.

Table 3 — Citation roles in the novelty and terminology boundary.

Literature family	Role in this manuscript
Bedian; Nieselt-Struwe-Wills; Füchslin-McCaskill; Wills-Nieselt-McCaskill; Wills/Carter	Establish that coding self-organization, self-description, assignment catalysts, physical interpreters, coding specificity, and reflexive coding are not standalone originality claims.
Vitas-Dobovišek; Chatterjee-Yadav; Spirov; Lei-de Farias-Burton	Establish prior chemical-historical, simulation, and sequence-reconstruction routes to translation and the Standard Genetic Code; distinguish those supplied or historically reconstructed molecular architectures from the present operational-intervention target.
Thlusty; Piraveenan-Polani-Prokopenko	Representative noisy-channel, information-theoretic, and fitness/selection frameworks; clarify the contrast with the present no-mapping-quality objective.
Görlich-Dittrich; Dittrich	Establish prior molecular-code and contingent-mapping/semantic-capacity formalization in reaction networks and physical media.
Rocha; Pattee; Sharov	Provide self-organization/symbol, modeling-relation, coenzyme-code, and interpretation-system contexts; delimit stronger notions of semantic closure.
Lacková-Faltýnek; Barbieri	Current code-threshold and Code Biology boundary; show that criteria for “code” extend beyond a single agreed definition.
Clark-Hickinbotham-Stepney	Establish semantic-closure precedent in artificial chemistry with executable architecture.
Nunes Palmeira et al.	Current comparator for heredity/coding emergence under protocell growth selection with copying/translation processes supplied.
Deacon et al.; Segura	Current conceptual/preprint boundaries for molecular-information emergence and reflexive/autopoietic closure.
Ameta et al.; Pavlinova et al.	Experimental/review boundary showing that autocatalytic RNA networks, compartments, and compositional heredity are insufficient as coding-specific evidence.

Data, code, and reproducibility availability

The supplementary materials contain two Python scripts for the representative reduced examples and the explicit 51-variable numerical compatibility example, together with software requirements and instructions for running the checks. The reaction families and channel counts are specified in Appendix E. These materials are provided for inspection in support of technical evaluation. Unless expressly stated otherwise, all rights are reserved, and no general permission is granted for redistribution, adaptation, incorporation into another project, or commercial use. Permission for additional use or for a separately arranged independent validation may be requested through the repository’s public issue tracker. Numerical trajectories, parameter sweeps, and the interactive simulator are not premises of any theorem or conclusion.

Author Contributions

Kai Liang formulated the originating research question, set the research aims and conceptual constraints, and determined research priorities, the timing of further mathematical review, and which proposed changes to the research approach and scope were pursued. He critically reviewed the manuscript’s conceptual framing, interpretation of results, and presentation, and requested substantive revisions. His feedback prompted a clearer distinction between assignment formation within a supplied architecture and the origin of decoding capabilities. He retained control over publication decisions. No other human contributor participated.

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Appendix A. Formal model, assay, and anti-circularity grammar

A physical model is written

$$\mathcal{M} = (\mathcal{S}, \mathcal{R}, N, a, \mu_0, \mathcal{B}, \mathcal{E}, \mathcal{L}),$$

with an analysis-only assay interface

$$\mathcal{A} = (\Sigma, \Omega, \phi_\Sigma, \phi_\Omega, \tau).$$

Deleting $\mathcal{A}$, the fitted mapping, mutual-information calculations, and all reporting code leaves the physical path law unchanged. STRICT-NC further requires independent assignment neutrality on declared degenerate classes, no decoder-equivalent primitive, no active initial interpreter, no hidden fitness/target score, and a declared material/energy boundary. Its typed grammar permits domain-local i -recognition and j -actuation; the only rule that first combines the two ports is the uniform $R_i + E_j \rightleftharpoons B_{ij}$ assembly. Every later joint-index rule consumes an already assembled carrier, and no direct $S_i \rightarrow O_j$ rule, primitive T_{ij} , or pair-conditioned rate is allowed.

The operational response is defined by challenge-dependent fluxes

$$J_t^\tau(\omega \mid \sigma) = E[N_\omega(t, t + \tau) \mid do(\sigma)]$$

and normalized channel K . For finite active alphabets,

$$C(K) = \max_{\sigma \neq \sigma'} TV(K(\cdot \mid \sigma), K(\cdot \mid \sigma')).$$

For a binary row-stochastic channel, $C(K) = |\det K|$. Mutual information is allowed only as a post-hoc diagnostic.

The interpreter criterion additionally requires a matched ablation that either destroys a required output channel by zeroing its total flux or, when the normalized ablated channel remains defined, strictly reduces the declared channel specificity/fidelity. No normalized K is defined at zero total output flux. Heredity and reflexivity are independent gates: heredity requires a parent-relative descendant intervention contrast; reflexivity requires matched blocking of code-mediated supporting output to reduce future interpreter capacity.

Exact finite-window readout calculation

Condition on a physical sampled state and carry out the preparation in Section 2.3. Write $\ell_{ij} = n_{A_{ij}} + n_{U_{ij}}$ for the initial accessible carrier count. For challenge i , each of those ℓ_{ij} carriers starts unbound and subsequently has the two-state transitions

$$A \xrightarrow{a_S} U, \quad U \xrightarrow{b_S} A, \quad U \xrightarrow{c_Q} A + O_j.$$

The last transition is marked as a new output event. Clamped symbol and readout activities make every carrier's rates first order, independent of volume and other carriers. Its bound probability $u(s)$ solves

$$u'(s) = a_S - (a_S + b_S + c_Q)u(s), \quad u(0) = 0.$$

Consequently $u(s) = a_S(1 - e^{-\lambda s})/\lambda_S$, and the compensator identity gives the expected number of marked firings per carrier as $c_Q \int_0^\tau u(s) ds = \Gamma_{\eta, \tau}$. Summing over carriers yields the formula in Section 2.3 exactly. For the other receptor type the challenge activity and initial bound count both vanish, so it contributes no output. The deterministic mass-action equations are linear and obey the identical integrated-flux identity. The expectation is finite, with $0 < \Gamma_{\eta, \tau} < c_Q \tau$. It is the **ratio of expected counts**, not an expectation of an empirical ratio with possibly empty rows.

At any finite count state with positive row totals, normalization cancels both $\Gamma_{\eta, \tau}$ and the volume. There is thus no missing finite-volume approximation in the descendant channel. Write $L_{i\bullet} = (L_{i0}, L_{i1})$ for a row vector. On a compact physical set where every row sum is at least $\ell_0 > 0$, the row-normalization inequality is

$$\text{TV} \left(\frac{L_{i\bullet}}{\|L_{i\bullet}\|_1}, \frac{\tilde{L}_{i\bullet}}{\|\tilde{L}_{i\bullet}\|_1} \right) \leq \frac{\|L_{i\bullet} - \tilde{L}_{i\bullet}\|_1}{\ell_0}.$$

For physical states $x, \tilde{x}$ in such a set, each $L_{ij} = A_{ij} + U_{ij}$ therefore gives

$$d_K(\mathcal{K}(x), \mathcal{K}(\tilde{x})) \leq \frac{4}{\ell_0} \|x - \tilde{x}\|_\infty.$$

Also $|C(K) - C(\tilde{K})| \leq 2d_K(K, \tilde{K})$ by the triangle inequality for total variation. These estimates justify the concentration-to-channel steps and the $O(\eta) + O(\nu)$ lifting when the limiting row totals are positive. A state with an empty required row is outside these channel neighborhoods and counts as failure in the lineage event, with no arbitrary channel value assigned. All controls are applied only in independent side samples; no measured channel or decoded assignment feeds back into the autonomous lineage law.

Appendix B. Null and minimal-relation proofs

For the identity-erasing null, aggregate the two challenged symbol identities before output selection. Theorem 1 assumes more than ordinary lumpability under each challenge: both intervention generators induce the same quotient generator, the projected initial laws coincide, and every marked-output intensity is the same function of the quotient state. With a well-posed nonexplosive common quotient problem and finite expected counts, the quotient process laws coincide. Integrating the common output-mark intensities against those laws gives equal expected output counts. The joint marked path laws are not asserted to coincide. Positivity of the total output flux makes both channel normalizations well-defined, so the rows are equal and $C(K) = 0$. Random trajectory-level output imbalance does not change this conclusion.

If a mediator preserves symbol identity but the output transition law, conditional initial law after each matched challenge, and matched boundary/intervention conditions are all invariant under a transitive output-label action, the same output-symmetry argument forces a uniform relative output law. A cross-domain relational degree of freedom is therefore necessary under these assumptions.

In the minimal binary factorized assembly, all four free $R_i - E_j$ edges have equal propensity. The first edge uniquely determines the second compatible edge. Two first choices lead to the diagonal perfect matching and two to the off-diagonal matching, yielding exact absorption probabilities $1/2$ and $1/2$. No target orientation is encoded in the rule grammar.

Appendix C. Symmetry breaking and stochastic branch selection

At a symmetric equilibrium of

$$\dot{x}_{ij} = [a + g(x_{ij})]r_i e_j - dx_{ij},$$

a checkerboard perturbation preserves row and column sums to first order. Its eigenvalue is

$$\lambda_A = g'(x)(1 - 2x)^2 - d = \frac{(1 - 2x)^2}{x} [xg'(x) - g(x) - a].$$

This gives Theorem 3 directly. For $g = bx^2$, the checkerboard-invariant reduction gives equations (8)–(9), and the full four-dimensional Jacobian decomposes into total/assignment and row/column sectors. The broken branch is linearly stable exactly for $u > 1/3$.

For finite N , output relabeling is an exact generator automorphism. The empty initial state is fixed by it and the two coding basin neighborhoods are exchanged. Irreducibility guarantees eventual hit of their union; path-law symmetry then gives exact first-hit probabilities $1/2$ and $1/2$. The neighborhoods in Section 4.4 are fixed before N , and their closures have positive distance δ_B from S_ρ . A hit therefore requires a fixed positive deviation from the exactly symmetric deterministic trajectory. The stopped concentration bound, including its first-exit buffer argument, proves that selection of these mature neighborhoods disappears if $N \rightarrow \infty$ is taken first at fixed time.

Appendix D. Reflexive-maintenance proofs

At a broken equilibrium, the damaged-state equation gives $D = cA$, with $c = \delta/(\ell + \rho y)$. Subtracting the two active equilibrium equations yields $pq = \alpha/b$, showing that the maintenance variable does not define the assignment. The implicit-function theorem at $\delta = 0$ continues the stable broken state to small positive damage.

For the support-block intervention, only the maintenance-production source changes at $t = 0$. The first derivatives of the active and damaged relation variables initially agree, while $\dot{y}$ differs by $-\beta A_{tot}^*$. Differentiating the active relation equation once more gives the t^2 capacity contrast in (24)–(25). In the reduced model, disabling the realized readout path removes the code-mediated maintenance-production flux at first order, giving (26). In the elementary finite- η assay, both arms first replace inherited O_j by equal-moiety inert isomers; the intact arm then produces new active output while the disabled arm produces only inert output. The resulting strict active-output contrast at order t and cumulative maintenance-production contrast at order t^2 give (26a), without counting pre-intervention $O_j = O(\eta)$ as newly decoded output.

Appendix E. Elementary CRN, conservation, and singular perturbation

The finite- η elementary process is fixed by the following **stated stochastic mass-action convention and complete schema table**. This is the concrete model used by Theorems 10, 14, 15, and 17; each fast reaction has an explicit convention.

For a reaction r with internal-state reactant multi-index α_r , volume $V_N(t)$, and coefficient $k_r(\eta, \theta)$ (with chemostat activities already folded into the coefficient), the count intensity is

$$\lambda_r(t, n) = V_N(t) k_r(\eta, \theta) \prod_s \frac{(n_s)_{\alpha_{rs}}}{V_N(t)^{\alpha_{rs}}},$$

where $(n)_m = n(n-1)\cdots(n-m+1)$. Thus the deterministic density rate is $k_r \prod_s x_s^{\alpha_{rs}}$. The deterministic volume protocol supplies the geometric dilution term $-gx$; dilution is not counted as a reaction channel. Chemostats/fuels/symbol challenges are boundary parameters and cumulative wastes are external sinks/bookkeeping coordinates.

For every $(i, j) \in \{0, 1\}^2$, the host pair schema is:

family	reaction	coefficient $k_r(\eta, \theta)$	instances
basal association	$R_i + E_j \rightarrow B_{ij}$	k_B^+/η	4
basal dissociation	$B_{ij} \rightarrow R_i + E_j$	k_B^-/η^2	4
activation	$B_{ij} + F_A \rightarrow A_{ij} + W_A$	$k_A F_A/\eta$	4
dimerization	$2A_{ij} \rightarrow H_{ij}$	k_H^+/η	4
dimer dissociation	$H_{ij} \rightarrow 2A_{ij}$	k_H^-/η^2	4
receptor capture	$H_{ij} + R_i \rightarrow X_{ij}$	k_R^+/η^2	4
receptor release	$X_{ij} \rightarrow H_{ij} + R_i$	k_R^-/η^2	4
effector capture	$X_{ij} + E_j \rightarrow Y_{ij}$	k_E^+/η^2	4
effector release	$Y_{ij} \rightarrow X_{ij} + E_j$	k_E^-/η^2	4
templating catalysis	$Y_{ij} + F_R \rightarrow H_{ij} + A_{ij} + W_R$	$k_{cat} F_R/\eta$	4
symbol binding	$A_{ij} + S_i \rightarrow U_{ij}$	$k_S^+ S_i/\eta$	4
symbol release	$U_{ij} \rightarrow A_{ij} + S_i$	k_S^-/η^2	4
readout	$U_{ij} + Q \rightarrow A_{ij} + S_i + O_j$	$k_Q Q/\eta$	4
damage	$A_{ij} \rightarrow D_{ij}$	δ	4
repair	$D_{ij} + M \rightarrow A_{ij} + W_M$	ρ	4
active turnover	$A_{ij} \rightarrow R_i + E_j$	d	4
damaged turnover	$D_{ij} \rightarrow R_i + E_j$	ℓ	4

The host-global reactions are

family	reaction	coefficient	instances
output-to-maintenance	$O_j + F_M \rightarrow M + W_j$	$k_M F_M/\eta$	2
maintenance decay	$M \rightarrow W_M^{(0)}$	μ	1

so the fixed-volume host inventory contains $17 \times 4 + 2 + 1 = 71$ instantiated channels.

For the growing model, add exactly the four code-blind material imports

family	internal-state reaction	count intensity	instances
receptor material import	$\emptyset \rightarrow R_i$	$gC_R^* V_N(t)$	2
effector material import	$\emptyset \rightarrow E_j$	$gC_E^* V_N(t)$	2

where the displayed zero-order form is the internal-state representation of the chemostatted precursor channels of Section 7.3.

Finally, for each (i, j) , the robust parasite/error extension is

family	reaction	coefficient	instances
label-blind error	$A_{ij} \rightarrow P_{ij}$	ν	4
parasite receptor capture	$P_{ij} + R_i \rightarrow X_{ij}^P$	$1/\eta$	4
parasite receptor release	$X_{ij}^P \rightarrow P_{ij} + R_i$	$1/\eta^2$	4
parasite effector capture	$X_{ij}^P + E_j \rightarrow Y_{ij}^P$	$1/\eta^2$	4
parasite effector release	$Y_{ij}^P \rightarrow X_{ij}^P + E_j$	$1/\eta^2$	4
parasite catalysis	$Y_{ij}^P + F_R \rightarrow 2P_{ij} + W_R$	κ/η	4
parasite damage	$P_{ij} \rightarrow Z_{ij}$	δ_P	4
parasite repair	$Z_{ij} + M \rightarrow P_{ij} + W_M$	ρ_P	4
parasite turnover	$P_{ij} \rightarrow R_i + E_j$	d_P	4
damaged-parasite turnover	$Z_{ij} \rightarrow R_i + E_j$	ℓ_P	4

which adds $10 \times 4 = 40$ channels. Hence the final growing robust process has exactly

$$71 + 4 + 40 = 115$$

reaction channels. The reaction families and their multiplicities are specified above.

All host and parasite structural coefficients not fixed by the displayed scaling are strictly positive on the admissible parameter region; the label-blind error coefficient ν may be zero or positive as stated in Section 8.2. In the assignment-neutral assay the boundary activities satisfy the declared type symmetries. Formal composition vectors described in Section 6 give zero net change for every declared conserved moiety in every internal reaction instance; only the four explicit growth-material imports alter receptor/effector totals.

To discharge normal hyperbolicity rather than assume it, scale the first-layer intermediates by $\bar{B} = B/\eta$, $\bar{H} = H/\eta$, $\bar{X} = X/\eta$, $\bar{Y} = Y/\eta$, and $\bar{U} = U/\eta$, and use fast time $\tau = t/\eta^2$. At $\eta = 0$, slow variables and free resources are parameters to leading order. The scalar $\bar{B}$ and $\bar{U}$ blocks have eigenvalues $-k_{\bar{B}}^-$ and $-k_{\bar{S}}^-$. The only nontrivial block, for each physical pair, is

$$J_{HXY} = \begin{pmatrix} -(a+b) & c & 0 \\ b & -(c+d) & e \\ 0 & d & -e \end{pmatrix},$$

where $a = k_H^- > 0$, $b = k_R^+ R_i \geq 0$, $c = k_R^- > 0$, $d = k_E^+ E_j \geq 0$, and $e = k_E^- > 0$. Its characteristic polynomial is $\lambda^3 + a_1 \lambda^2 + a_2 \lambda + a_3$, with

$$a_1 = a + b + c + d + e, \quad a_2 = ac + ad + ae + bd + be + ce, \quad a_3 = ace.$$

All three coefficients are positive, and the remaining cubic Routh–Hurwitz margin [Gantmacher 1959] expands as

$$\begin{aligned} a_1 a_2 - a_3 &= a^2 c + a^2 d + a^2 e + abc + 2abd + 2abe + ac^2 + 2acd + 2ace \\ &\quad + ad^2 + 2ade + ae^2 + b^2 d + b^2 e + bcd + 2bce + bd^2 + 2bde \\ &\quad + be^2 + c^2 e + cde + ce^2 > 0. \end{aligned}$$

Hence every first-layer fast eigenvalue has strictly negative real part for every physical $R_i, E_j \geq 0$. The critical manifold is unique and smooth, with

$$\begin{aligned}\bar{B} &= K_B R_i E_j, & \bar{H} &= K_H A_{ij}^2, \\ \bar{X} &= K_R K_H A_{ij}^2 R_i, & \bar{Y} &= K_E K_R K_H A_{ij}^2 R_i E_j, \\ \bar{U} &= K_S A_{ij} S_i.\end{aligned}$$

Here $K_B = k_B^+/k_B^-$, $K_H = k_H^+/k_H^-$, $K_R = k_R^+/k_R^-$, $K_E = k_E^+/k_E^-$, and $K_S = k_S^+/k_S^-$ under the scaled convention in the reaction table. Substitution into the active-unit and output balances gives exactly the three induced slow coefficients (39)–(39a). The constructive choices (39b) therefore map any selected positive reduced (α, b, β) to a strictly positive elementary witness; ordinary continuity of these formulas maps an open microscopic neighborhood of that witness into a neighborhood of the selected reduced triple.

On any compact slow core, spectral continuity gives a uniform first-layer gap. For the unit-equilibrium-constant microscopic witness with $0 \leq R, E \leq 1$, shifting the cubic by 0.1 leaves all Routh–Hurwitz inequalities strict, so one may take the explicit conservative bound $\max \Re \lambda < -0.1$.

To identify the actual slow host variable, define

$$a_{ij} = A_{ij} + U_{ij} + 2(H_{ij} + X_{ij} + Y_{ij}).$$

Writing the elementary reaction fluxes explicitly and differentiating this combination shows exact cancellation of every singular association, dissociation, and readout-cycle flux. Before growth is added,

$$\dot{a}_{ij} = \frac{k_A F_A}{\eta} B_{ij} + \frac{k_{cat} F_R}{\eta} Y_{ij} - (d + \delta) A_{ij} + \rho M D_{ij}.$$

The damaged coordinate satisfies $\dot{D}_{ij} = \delta A_{ij} - (\ell + \rho M) D_{ij}$. On the fast manifold, $A = a + O(\eta)$, $B/\eta = K_B R_i E_j + O(\eta)$, and $Y/\eta = K_E K_R K_H A^2 R_i E_j + O(\eta)$. Moreover the exact receptor material bound differs from $a + D$ by $B + X + Y = O(\eta)$, and the exact effector bound differs from $a + D$ by $B + Y = O(\eta)$. Hence the free-resource expressions and the (a, D, M) slow field equal the reduced Section 5 field plus $O(\eta)$. With neutral growth, common dilution adds $-g(a, D, M)$; the code-blind growth-material boundary acts only on the total-moiety coordinates. This proves the host slow-coordinate lifting required by Theorem 10 and Theorem 14 and explicitly rules out any claim that the full finite- η internal field is exactly the reduced field.

After the first layer is slaved, set $\bar{O} = O/\eta$ and use the second fast time t/η . The leading output block has eigenvalue $-k_M F_M < 0$, uniformly bounded away from zero when F_M stays in the declared compact positive range. Thus both nested critical manifolds are normally hyperbolic and attracting as a consequence of the concrete grammar. Nested Tikhonov–Fenichel persistence [Tikhonov 1952; Fenichel 1979] then gives a locally invariant slow manifold on which the full slow vector field in the active-unit coordinates is the reduced Section 5 field plus $O(\eta)$. At $g = 0$, stability is relative to the fixed receptor/effector stoichiometric compatibility class; the four ambient conserved-total directions are zero modes and are not counted as unstable directions. For $g > 0$, the four code-blind material-import channels replace them by the exact transverse modes $-g$. Hyperbolic equilibria and strict causal inequalities are structurally stable under this perturbation. For intervention (26a), the common initial replacement makes active $O_j(0) = 0$ in both arms. Taylor expansion of the exact output balance gives $O_{intact}(t) - O_{disabled}(t) = r_O^* t + O(t^2)$, and integration of the output-to-maintenance propensity gives the positive coefficient displayed in (26a). Hence the finite- η causal contrast is an exact consequence of the elementary equations, not an inference from the reduced zero-output idealization.

For clarity, collect the scaled host intermediates into $v = (B, H, X, Y, U)/\eta$, the scaled outputs into $w = (O_0, O_1)/\eta$, and the active-unit, damage, maintenance, and permitted material coordinates into z . On the selected local compact set the physical-time system has the form

$$\eta^2 \dot{v} = F(v, w, z; \eta), \quad \eta \dot{w} = G(v, w, z; \eta), \quad \dot{z} = H(v, w, z; \eta),$$

with smooth right-hand sides after the coordinate substitution. The first critical graph is obtained from the attracting B, U, HXY blocks computed above. After its elimination, the output layer has eigenvalues $-k_M F_M < 0$. Both critical graphs are smooth and locally uniformly attracting. The equilibrium equations can therefore be solved successively by the implicit function theorem. Successive Schur complements

separate the two fast spectral groups from the reduced slow spectrum, preserving their strict negative real parts for sufficiently small positive η .

For fixed-time trajectory convergence, the two attracting layers are used in this order, with scale ratio $\eta^2/\eta \rightarrow 0$, initial data in their attraction domains, and the reduced trajectory remaining in the selected local domain [Tikhonov 1952, pp. 576–577, Theorem 2]. Smooth normal hyperbolicity gives the local C^1 persistence used in the slow-field comparison [Fenichel 1979]. The fixed-time maintenance comparison chooses its positive observation time before taking η small. At $g = 0$, the slow stability statement is restricted to the fixed material compatibility class.

Appendix F. Heredity concentration, information cuts, and resets

Equation (44) is a direct union of two Hoeffding bounds over the $D_{\text{host}} = 35$ host-only internal coordinates in the pure-partition assay. Continuity of the channel map on a compact basin core converts concentration fidelity into channel fidelity.

For an information cut containing m ancestral copies, write $Z_H = \sum_k I_{k,H}$. The marginal bound $\Pr(I_{k,H} = 1) \leq q_H = (\varrho/2)^H$ gives $\Pr(Z_H \geq 1) \leq \min\{1, mq_H\}$ by the union bound, without any cross-copy independence assumption. Conditioning on $Z_H \geq 1$ versus $Z_H = 0$ gives (49), and letting $H \rightarrow \infty$ proves Theorem 13. Only when the copy histories and their edge trials satisfy the additional independence conditions stated in Theorem 12 does $Z_H \sim \text{Bin}(m, q_H)$, yielding (49a) and the copy-number threshold (50). Formula (51) additionally requires independence across cuts and the joint failure baseline stated there; individual cut-loss baselines alone do not suffice.

For the long-lineage theorem, fix finite positive η, g , a parameter vector θ , an orientation σ , and the compact stopped region $K^\sigma(\theta)$. During a growth cycle write $v(t) = e^{gt}$, $V_N(t) = Nv(t)$, and $X^N(t) = Z^N(t)/(Nv(t))$. Exact stochastic mass action gives

$$Z^N(t) = Z^N(0) + \sum_r Y_r \left(\int_0^t Nv(s) a_r^{N,v(s)}(X^N(s)) ds \right) \nu_r,$$

where the falling-factorial density rates $a_r^{N,v}$ converge uniformly on K^σ to the continuous mass-action rates a_r with error $O(1/N)$. The four explicit growth-material boundary channels have count intensities $gC_R^* Nv(t)$ and $gC_E^* Nv(t)$, hence exact constant density rates and no falling-factorial correction; they are included in the reaction index r , in F^N, F , and in the stopped tube. Integration by parts therefore yields

$$X^N(t) = X^N(0) + \int_0^t (F^N(s, X^N(s)) - gX^N(s)) ds + M^N(t),$$

with compensated jumps $\nu_r/(Nv(t))$. Make the stopping explicit by setting

$$\tau_K^N = \inf\{t \geq 0 : X^N(t) \notin K^\sigma\}, \quad M^{N,K}(t) = M^N(t \wedge \tau_K^N).$$

All rate bounds below apply to physical states before this exit, including the predictable pre-jump state of the exit jump. Let $F_g = F - gx$ be L_g -Lipschitz on the compact region containing the stopped states and deterministic path. Let B_F/N bound $\|F^N - F\|_\infty$, and choose uniform upper bounds $\bar{a}_r$ on the exact density rates $a_r^{N,v}$ there for all sufficiently large N . Define

$$J = \max_r \|\nu_r\|_\infty, \quad V_* = \max_i \sum_r \nu_{r,i}^2 \bar{a}_r, \quad \Xi_g(T) = \int_0^T e^{-gs} ds.$$

Because $v(t) \geq 1$, each stopped martingale jump is at most J/N , and its predictable quadratic variation satisfies

$$\langle M_i^{N,K} \rangle_T \leq \frac{V_* \Xi_g(T)}{N}.$$

Freedman's maximal inequality [Freedman 1975] therefore gives

$$\Pr\left(\sup_{t \leq T} |M_i^{N,K}(t)| > z\right) \leq 2 \exp\left[-\frac{Nz^2}{2(V_*\Xi_g(T) + Jz/3)}\right].$$

The Gronwall inequality [Gronwall 1919], applied up to the stopping time, gives

$$\sup_{t \leq T \wedge \tau_K^N} \|X^N(t) - \phi_t(x_0)\|_\infty \leq e^{L_g T} \left(\sup_{t \leq T} \|M^{N,K}(t)\|_\infty + \frac{B_F T}{N} \right).$$

Fix the deterministic path and a positive ambient buffer

$$\delta_K = \inf_{t \leq T} \text{dist}_\infty(\phi_t(x_0), (K^\sigma)^c) > 0.$$

For $0 < \varepsilon < \delta_K$, choose N_0 so that $B_F T/N \leq \varepsilon e^{-L_g T}/2$, and set $z = \varepsilon e^{-L_g T}/2$. A union bound over coordinates yields

$$\Pr\left(\sup_{t \leq T \wedge \tau_K^N} \|X^N(t) - \phi_t(x_0)\|_\infty > \varepsilon\right) \leq 2D_{state} e^{-Nc_M},$$

where

$$c_M = \frac{z^2}{2(V_*\Xi_g(T) + Jz/3)} > 0.$$

On a first exit by time T , the stopped trajectory deviation is at least $\delta_K > \varepsilon$. If no exit occurs, the stopped and unstopped deviations on $[0, T]$ agree. Therefore the event that the unstopped deviation exceeds ε is contained in the stopped deviation event, and it obeys the same bound. Lemma 5 supplies an ambient tube with a common positive buffer for all permitted starts in $D_{out}^\sigma(\theta)$, making this estimate uniform over those starts. All parameters, T , and tube margins are fixed before taking N large. The same argument applies at $g = 0$, with $v(t) = 1$ and $\Xi_0(T) = T$, to the fixed-volume process in Theorem 6; its state dimension replaces D_{state} .

At T_g , the parent volume is $2N$. Independent Bernoulli $1/2$ molecule partition into a tracked daughter of reset volume N gives the conservative Hoeffding estimate used in Section 7.4. Choosing $D_{in}^\sigma(\theta) \subseteq D_{out}^\sigma(\theta) \subseteq K^\sigma(\theta)$, applying the maturation bound uniformly on the stopped tube, and then the fission bound gives (58) with $C_0 = 4D_{state} = 204$. Because the estimate is uniform for every $\mathcal{F}_h$ -measurable start in D_{out}^σ , conditional iteration and a union bound give (59)–(60).

For the host-only matched causal comparator, the topology-erasure intervention is the explicit product rewiring kernel $\mathcal{E}$ defined before Theorem 11 over $\mathcal{C}_{pair}^{host}$. For the final positive-error/free-rider model, use the extension $\mathcal{E}_{full}$ of Section 8.2 over $\mathcal{C}_{pair}^{full} = \{B, A, D, U, H, X, Y, P, Z, X^P, Y^P\}$. In each included structural class, receptor-side and effector-side object slots are uniformly re-matched while the declared balanced weighted marginals, class totals, bulk coordinates, and burden are fixed. The relevant kernel satisfies the same equivariance identity, $\mathcal{E}_*(\rho x, \rho A) = \mathcal{E}_*(x, A)$, where $\mathcal{E}_*$ denotes $\mathcal{E}$ or $\mathcal{E}_{full}$ according to the model state. On the balanced mature parent the retained statistic is ρ -invariant, hence the erased-parent law is ρ -invariant; subsequent chemistry, growth, and single-lineage fission are equivariant. Each specified disjoint orientation neighborhood therefore has erased probability at most $1/2$, yielding (61).

For Theorem 16, either its abstract reset hypotheses or the no-reaction/empty-daughter construction (63) gives $\Pr(\tau_{exit}^{N,\sigma} > h + 1 \mid \mathcal{F}_h, \tau_{exit}^{N,\sigma} > h) \leq 1 - p_N$ uniformly in the surviving history. Induction yields the product bound in (62). This proof fixes η, g and the θ -dependent compact basin geometry before taking N large; it does not claim a coupled singular/stochastic limit.

Appendix G. Finite- η free-rider lifting and physical error

Linearizing the reduced parasite active/damaged subsystem gives (64). For a 2×2 Metzler matrix with negative second diagonal term, the dominant eigenvalue is negative exactly when the determinant is positive together with the negative trace condition. Algebraic reduction yields (66). Under matched host/parasite damage-repair parameters, substituting the host equilibrium equation collapses the **reduced** threshold to $\kappa_c = bu_g$.

For the explicit elementary parasite grammar, use the slow active-unit coordinate $p = P + X^P + Y^P$ and fast coordinates $X^P = \eta\bar{X}$, $Y^P = \eta\bar{Y}$. On fast time $\tau = t/\eta^2$, the leading fast system is

$$\frac{d}{d\tau} \begin{pmatrix} \bar{X} \\ \bar{Y} \end{pmatrix} = \begin{pmatrix} pR - (1+E)\bar{X} + \bar{Y} \\ E\bar{X} - \bar{Y} \end{pmatrix} + O(\eta).$$

Its fast Jacobian has trace $-(2+E)$ and determinant 1, so it is uniformly Hurwitz on each compact physical resource range. The unique leading critical manifold is $\bar{X} = pR$, $\bar{Y} = pRE$. Summing the P, X^P, Y^P equations cancels every binding/unbinding flux and gives on that manifold

$$\dot{p} = [\kappa RE - (d_P + \delta_P + g)]p + \rho_P MZ, \quad \dot{Z} = \delta_P p - (\ell_P + \rho_P M + g)Z,$$

which is exactly (64) at $R = E = s_g$, $M = y_g$. Fix the reduced macro parameters and $g \in (0, g_*)$, and take the finite- η host family on $0 < \eta < \eta_H(g)$. If the reduced block satisfies $\Lambda_P^{\text{red}}(g) \leq -m_P < 0$, normal-hyperbolic persistence and spectral continuity give $\eta_P(g) > 0$ such that the finite- η slow invasion Jacobian is $L_P^{\text{red}}(g) + O(\eta)$ and remains strictly stable for every $0 < \eta < \min\{\eta_H(g), \eta_P(g)\}$. This is the analytic elementary lifting. The exact equality $\kappa_c = bu_g$ is retained only as the reduced/effective-model identity.

At $\nu = 0$, after ordering host variables before parasite variables, the full Jacobian is block triangular at the parasite-free state: parasite-row derivatives with respect to host perturbations vanish because all such coefficients multiply a parasite coordinate. The host block is hyperbolically stable and the finite- η parasite block D_η is strictly stable. Ordered by pairs, $D_\eta = \text{diag}_{ij} D_{\eta,ij}$ has four independent 4×4 blocks. Its off-diagonal entries are nonnegative transfer rates. Within each pair, the bidirectional links $P \leftrightarrow Z$, $P \leftrightarrow X^P$, and $X^P \leftrightarrow Y^P$ are positive at the host equilibrium, so each pair block is irreducible; there are no links between distinct pairs. A small uniform mutation $A_{ij} \rightarrow P_{ij}$ is a smooth perturbation with source vector $s_{ij} = (A_{ij}^*, 0, 0, 0)^T$ in every block. For the IFT branch $q(\nu)$ of parasite coordinates,

$$q'(0) = -D_\eta^{-1}s, \quad -D_\eta^{-1} = \int_0^\infty e^{D_\eta t} dt \geq 0.$$

For each irreducible Hurwitz Metzler block, positive-linear-system theory [Farina and Rinaldi 2000] gives $e^{D_{\eta,ij}t} > 0$ for $t > 0$. Since each s_{ij} has a positive entry, the block integral $\int_0^\infty e^{D_{\eta,ij}t} s_{ij} dt$ is strictly positive in all four coordinates. Applying this argument to all four blocks yields $q'(0) > 0$ componentwise without an irreducibility claim for the full matrix. Consequently, for each fixed admissible η , there exists $\nu_0(g, \eta) > 0$ such that $0 < \nu < \nu_0$ gives a physical strictly positive parasite branch with $P, Z, X^P, Y^P = O(\nu)$; $\nu = 0$ gives the parasite-free boundary point. The positive host/free-resource coordinates and all strict coding and reflexive-maintenance margins persist by continuity.

The robust-state topology-erasure kernel must include P, Z, X^P, Y^P as well as the host pair-bearing classes. Classwise uniform slot rewiring preserves the corresponding weighted material marginals and is output-swap equivariant. The small-error robust checkerboard continuation remains in the simultaneous row/column-swap fixed subspace, so its preserved class margins are output-swap balanced. Hence the full erased robust-parent law is output-swap invariant and the named-orientation comparator ceiling remains $1/2$.

Neutral volume growth and unbiased fission cancel at the expectation level over a balanced doubling cycle, producing the expected linearized multiplier (69) for the reduced rare-parasite mode; this is not an almost-sure single-lineage multiplier. The finite- η elementary first-moment invasion exponent remains negative for sufficiently small η when the reduced margin is strict; equivalently, its cycle multiplier remains strictly below one.

Appendix H. Empirical nulls and matched-intervention margins

A composition-sufficient comparator is defined by a Markov kernel or deterministic transition law that factors through S_{comp} . Because the topology-erasure intervention leaves S_{comp} invariant, the intact and erased future laws are identical, proving (72).

For imperfect matching, the null class can move by at most $L_j\epsilon_j$ while the model lower bound can fall to $m_j - K_j\epsilon_j$. Strict inequality (74) leaves a nonempty interval between them. Its midpoint is (75), and the distance from that midpoint to either bound is the common separation margin (76).

Appendix I. Computational cross-checks

The accompanying Python scripts reproduce the reduced growth-equilibrium points in Figure 5 and the 51-dimensional numerical compatibility example in Section 8.4. Each script specifies its model parameters and reports numerical diagnostics. Instructions and software requirements are provided in the accompanying README.

Numerical calculations provide certificates or implementation checks unless a result is explicitly labelled computational. They are not substituted for the corresponding analytic theorem.

Appendix J. Literature and novelty boundary matrix

The literature boundary is organized by explanatory role rather than by superficial vocabulary. Prior work already covers assignment catalysts and primitive-code models; evolutionary self-organization of coding; physical and self-generating interpreters; reflexive coding; semantic closure in artificial chemistries; structural molecular-code capacity in reaction networks; autocatalytic/compositional heredity; and protocell evolution with predeclared copying/translation processes.

The originality claim is tied to a specific combination: relation formation from the declared initially unwired state, operational contingent readout, local non-score causality, parent-relative causal channel heredity, output-mediated maintenance, and the stated stochastic/error/free-rider robustness conditions in one supplied reaction architecture. The candidate addition is this coupled construction and its causal discrimination, not the first recognition that interpreters require explanation. No exhaustive priority or universal architectural minimality claim is made.

For meaningful comparison, a prior model should be assessed by its initial physical contents, recognition and conversion operations, source of any assignment preference, definition of the measured channel, lineage comparator, maintenance dependence, and joint parameter conditions. Sharing words such as coding or closure establishes neither equivalence nor difference. Section 10.6 gives focused comparisons; the full interpretive and mathematical assumptions determine the scope of the contribution.

If a prior construction is shown to meet the conjunction at equal or greater generality, the originality statement must be revised. Such a literature collision would change the framing, not retrospectively invalidate the mathematical results.